

# Nyahmation

User Manual

Version 0.0.1 · second edition: the camera, effects, walk cycles and automatic lip sync · September 2026

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**About the pictures.** The screenshots in this manual are of the real app. Numbered pink circles point at the parts of the screen that the list under each picture explains. Keyboard shortcuts are shown for a Mac: on Windows, use Ctrl wherever this manual says ⌘, and Alt for ⌥ Option.

THIS EDITION

# What's new

Everything added since the first version of this manual

The first edition of this manual covered drawing, rigging, posing, sound, lip sync and export. Since then Nyahmation has gained a camera, effects, tools for walks and repeats, and a lot of smaller helpers. If you already know the app, this page is a map of what to read. Chapters marked **new** in the contents are new from start to finish.

| WHAT | IN SHORT | READ |
|---|---|---|
| **The camera** | Pan, zoom and turn the view on the timeline with the Camera tool. The characters don't move; only what you see does. | Chapter 11 |
| **Depth and scrolling** | Far hills that move less than the foreground, a sky gradient, a title fixed to the screen, a speech bubble that follows a head, and scenery that scrolls past forever. | Chapter 12 |
| **Gradients** | Any shape can be filled with a straight or round gradient. | Chapter 4 |
| **Effects** | Drop shadow, outer glow, blur, haze and contact shadow; blend modes; clipping (pupils that stay inside the eye). All of the numbers can be animated. | Chapter 13 |
| **Cut, copy and paste** | Parts and whole layers in Build mode, even between projects; poses in Animate mode. | Chapter 4 |
| **Shape tools** | Union, Subtract, Intersect and Exclude combine shapes into new ones. | Chapter 4 |
| **Swatches and eyedropper** | A palette of the project's colours, and picking a colour from anywhere on the screen. | Chapter 4 |
| **Walks and repeats** | Mirror Pose, Swap Sides, and cycles: a layer that repeats a stretch of its animation, optionally moving on each time. | Chapter 10 |
| **Timing tools** | A custom easing curve, stretching a range of poses, draw-order changes over time, and switching between ones and twos mid-scene. | Chapter 9 |
| **Automatic lip sync** | Nyahmation listens to the dialogue and fills in the mouth shapes for you to polish. | Chapter 15 |
| **More export formats** | H.265, WebM (optionally see-through) and ProRes 4444 MOV for video editors. | Chapter 16 |
| **Library updates** | Save a new version of a character and update the projects that use it, keeping their animation. | Chapter 7 |
| **Autosave** | Unsaved work is kept safe and offered back if Nyahmation closes unexpectedly. | Chapter 17 |
| **Smooth preview** | A Preview quality setting, a live frame-rate readout, and a speed test for your computer. | Chapter 8 |

CHAPTER 1

# Welcome

What Nyahmation does, and the big idea

Nyahmation is a studio for making cartoons. You draw a character (or import one), connect its parts at joints like a paper puppet, pose it on the frames that matter, and Nyahmation fills in every frame in between. Add a voice, let Nyahmation (or you) match the mouth shapes to it, point a camera at the scene, and export a video you can play anywhere.

## The big idea: poses, not drawings

Think of a flip book. Drawing every page by hand is a lot of work. In Nyahmation you only set up the important pages, called **poses**: the arm down on frame 1, the arm up on frame 12. Nyahmation draws the 10 pages in between for you, smoothly. Want the move faster? Slide the second pose closer to the first. That's the whole trick, and the rest of this manual builds on it.

There is no difference between a "keyframe" and a normal frame: you can pose on any frame at any time, and only the parts you actually move get a pose. If you wave a hand on frame 20, the legs aren't affected.

## A puppet on a stage, filmed by a camera

It helps to picture three separate things, because Nyahmation keeps them separate too:

- **The puppets** (characters) and **the scenery** (backgrounds) stand on a stage. Posing moves them around the stage.
- **The camera** films the stage. Moving the camera changes what ends up in the picture, but never moves a puppet. A foot pinned to the floor stays pinned however the camera swoops.
- **The soundtrack** runs alongside. Nothing you do to the picture ever shifts the sound, so the lip sync you match to it stays matched.

When something surprises you, asking "am I moving the puppet, the camera, or the sound?" usually explains it.

> **Try it first.** Choose **File → Open Demo Puppet**. It opens Pip, a ready-made character who waves and talks. Poke around: you can't break anything, and ⌘ + Z undoes any change. Have fun, and make something that makes someone laugh.

## Three steps to a cartoon

1. **Build.** Draw your character as separate parts (head, body, arms, legs) and set up where each part bends. This is **Build** mode.
2. **Animate.** Switch to **Animate** mode, pick a frame, and pose. Add sound and lip sync, and move the camera.

3. **Export.** Save it as a video, with the sound.

CHAPTER 2

# Getting started

Installing, opening, and the demo puppet

## Opening Nyahmation

Once Nyahmation is installed in Applications, open it like any other app: from Launchpad, the Dock, or by double-clicking it. Double-clicking a saved project (a **.nyah** file) opens it too, and **File → Open Recent** lists the last ten projects you opened or saved.

A new scene starts empty, with a small card saying how to begin: draw with the tools on the left, drag pictures in from your computer, or open **Pip, the demo puppet**, to see how animating works. It goes away as soon as you draw something.

## Installing it on a Mac (for grown-ups)

Nyahmation isn't in the App Store; you make the app from its source code once. You need **Node.js** (the LTS version from nodejs.org) and a copy of the code (from GitHub: pick the branch, then **Code → Download ZIP**). Then, in Terminal, inside the Nyahmation folder:

```
npm install
npm run dist:mac
```

After 5–10 minutes a Finder window shows **Nyahmation-0.0.1.dmg**. Double-click it and drag Nyahmation onto Applications. That's it: from now on nobody needs Terminal.

If macOS blocks it the first time ("can't check it for malicious software"), click **Done**, then open **System Settings → Privacy & Security**, scroll down, and click **Open Anyway**. This happens because the app isn't signed with a paid Apple developer account; it's only needed once.

To try it without installing, run `npm run dev` in the same folder instead. The app opens and runs until you close Terminal.

### Automatic lip sync needs one extra download

**npm install** also fetches **Rhubarb Lip Sync**, the free program that listens to dialogue for Auto lip sync (Chapter 15). It's about 90 MB and needs the internet once. If that download was skipped or failed, run `npm run setup` in the Nyahmation folder, and build the app again. Everything else works without it. On a Mac with Apple Silicon, Rhubarb runs through Rosetta; if macOS offers to install Rosetta, say yes.

## Several people on one Mac

Apps in Applications are shared by every account on the Mac, so each person finds Nyahmation in their own Launchpad. Each person's projects and library stay in their own account. The first time someone saves to the library, macOS asks whether Nyahmation may use their Documents folder: click **Allow**.

## What you need

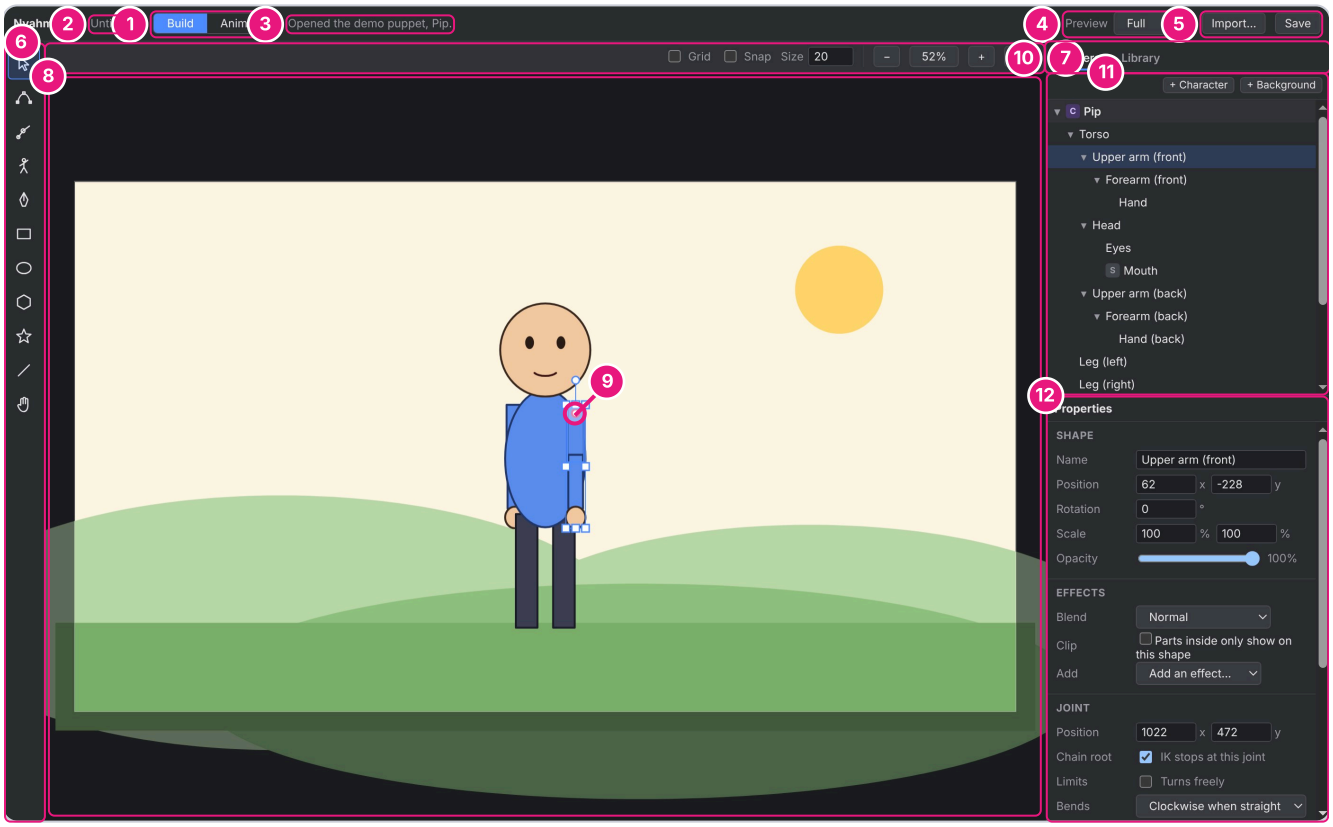
- A Mac with a recent version of macOS (Intel or Apple Silicon), or a Windows 10/11 PC. A 2020 MacBook Air is plenty.
- No internet connection, account or subscription. Everything stays on your computer.

CHAPTER 3

# The workspace

A tour of the window and the menus

Here is the whole window, with Pip the demo puppet open in Build mode and one part selected.




*The Nyahmation window in Build mode.*

1. Mode switch: Build (draw and rig) or Animate (pose on the timeline)
2. Project name; a dot means there are unsaved changes
3. Status: what just happened, and tips
4. Preview quality: Full, Half or Quarter (see Chapter 8)
5. Import art or sound, and Save
6. Toolbar: the drawing, rigging and view tools
7. Tool options, grid and snap, and zoom
8. The canvas: the light area is the scene (what gets exported)
9. Selected part, with its joint and handles to scale and rotate it
10. Panel tabs: Layers and Library
11. Layers panel: every layer and part, front to back
12. Properties of whatever is selected

## Two modes

| MODE | WHAT IT'S FOR |
|---|---|
| **Build** | Drawing, arranging and rigging. Changes here are the character's *resting* look: how it looks when nothing is animated. |
| **Animate** | Posing on the timeline, the camera, sound and lip sync. Every change you make is recorded as a pose on the current frame. The timeline appears at the bottom. |

## The menus

On a Mac the menus are in the menu bar at the top of the screen.

| FILE | |
|---|---|
| New | ⌘ + N |
| Open... | ⌘ + O |
| Open Recent | |
| Open Demo Puppet | |
| Save | ⌘ + S |
| Save As... | ⇧ + ⌘ + S |
| Import Art or Sound... | ⌘ + I |
| Record Dialogue... | ⇧ + ⌘ + R |
| Export Video... | ⌘ + E |

| EDIT | |
|---|---|
| Undo | ⌘ + Z |
| Redo (Windows: Ctrl + Y) | ⇧ + ⌘ + Z |
| Cut | ⌘ + X |
| Copy | ⌘ + C |
| Paste | ⌘ + V |
| Duplicate | ⌘ + D |
| Select All | ⌘ + A |
| Deselect | ⇧ + ⌘ + A |

| OBJECT | |
|---|---|
| Group | ⌘ + G |
| Ungroup | ⇧ + ⌘ + G |
| Combine Shapes | ⌘ + 8 |
| Shape Tools → Union | ⌥ + ⌘ + U |
| Shape Tools → Subtract | ⌥ + ⌘ + S |
| Shape Tools → Intersect | ⌥ + ⌘ + K |
| Shape Tools → Exclude | ⌥ + ⌘ + E |
| Make Switch Layer (Mouth, Eyes...) | ⇧ + ⌘ + M |
| Bring Forward | ⌘ + ] |
| Send Backward | ⌘ + [ |
| Bring to Front | ⇧ + ⌘ + ] |
| Send to Back | ⇧ + ⌘ + [ |
| New Character Layer | |
| New Background Layer | |
| Mirror Pose (flip left and right) | ⌥ + ⌘ + M |
| Swap Sides (for walks) | ⌥ + ⇧ + ⌘ + M |
| Mark Branch Joints as Chain Roots | |
| Save Selection to Library... | ⇧ + ⌘ + L |

| VIEW | |
|---|---|
| Zoom In | ⌘ + = |
| Zoom Out | ⌘ + − |
| Fit Scene | ⌘ + 0 |
| Actual Size | ⌘ + 1 |
| Show Grid | ⌘ + ' |
| Snap to Grid | ⇧ + ⌘ + ' |
| Measure Preview Speed | |

| HELP | |
|---|---|
| Nyahmation User Manual | |
| Keyboard Shortcuts (every shortcut on one sheet) | ⌘ + / |

The strip above the canvas reminds you how the current tool works, including what Shift, ⌥ and ⌘ do with it.

## Moving around the canvas

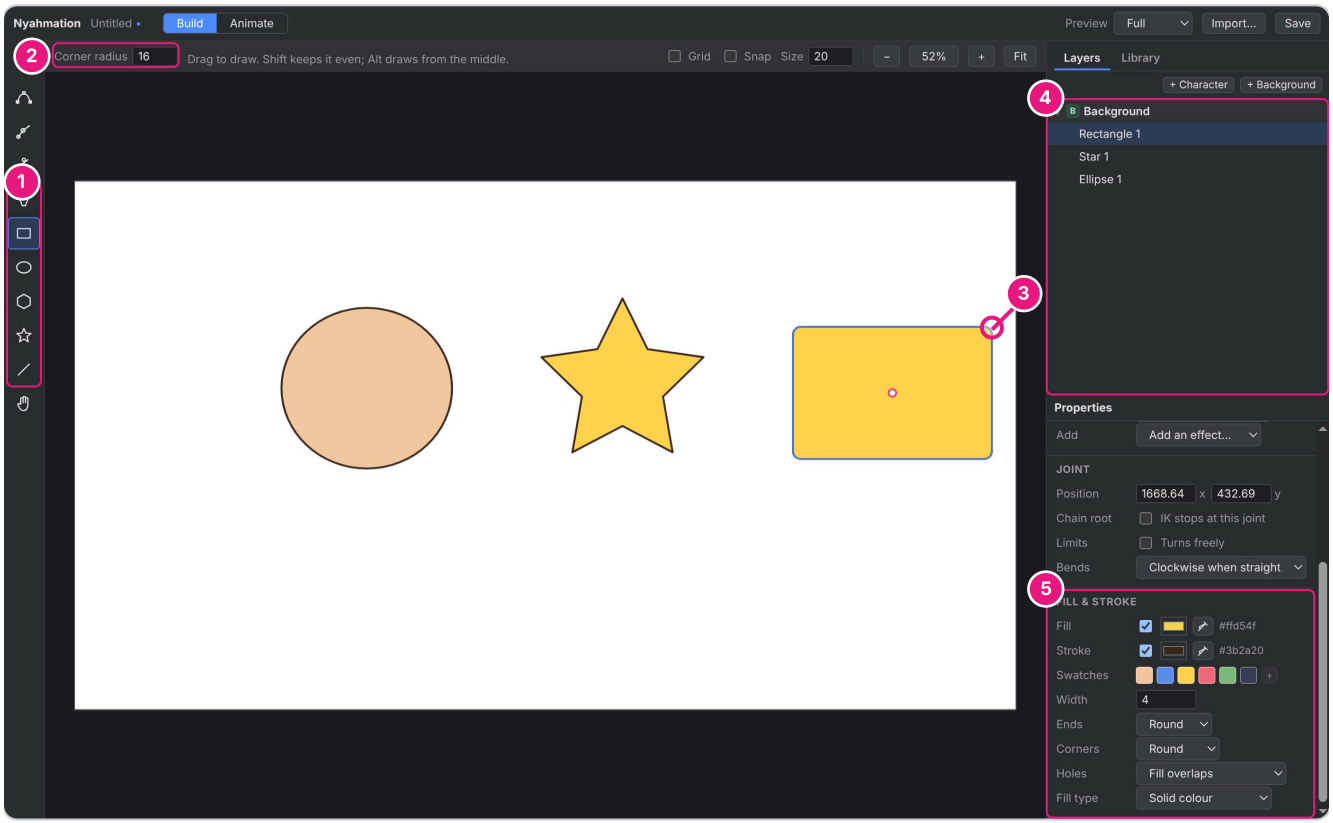
- **Pan:** scroll, or hold Space and drag, or use the Hand tool (H).
- **Zoom:** pinch on the trackpad, or hold ⌘ and scroll. **Fit** shows the whole scene.
- The light rectangle is the scene: exactly what ends up in the video. You can draw outside it, but that won't be exported (unless the camera looks there; see Chapter 11).

CHAPTER 4

# Drawing

Shapes, points, colours, gradients and importing art

Everything in Nyahmation is drawn with **vector** shapes: outlines made of points and curves. They stay perfectly sharp at any size, and you can reshape them at any time.




*Three shapes drawn with the Ellipse, Star and Rectangle tools. The rectangle is selected.*

1. Shape tools: Pen, Rectangle, Ellipse, Polygon, Star and Line
2. Options for the current tool (here, the rectangle's corner radius)
3. The new shape is selected: drag the handles to resize, the round one to rotate
4. Each shape appears in the Layers panel, inside the active layer
5. Fill and stroke: colours with eyedroppers, swatches, width, line ends, corners, holes and fill type

## The drawing tools

| TOOL | KEY | HOW TO USE IT |
|---|---|---|
| **Select** | V | Click a part to select it, and drag to move it. Drag the square handles to resize (⇧ keeps the proportions, ⌥ resizes from the centre) and the round handle to rotate (⇧ snaps to 15°). Drag on an empty spot to select everything in a box. Double-click a shape to edit its points. |
| **Points** | A | Reshape a shape by moving its points, their handles, or the curves between them. |
| **Pen** | P | Click to place corner points; click and drag to make curves. Click the first point to close the shape. Enter or a double-click finishes an open line; Backspace removes the last point. |
| **Rectangle** | M | Drag out a rectangle. Set rounded corners in the tool options. |
| **Ellipse** | L | Drag out an oval. Hold ⇧ for a circle. |
| **Polygon** | Y | Drag out a shape with equal sides. Choose how many sides in the tool options. |
| **Star** | S | Drag out a star. Choose the number of points and how deep the dents are. |
| **Line** | \ | Drag a straight line. ⇧ keeps it at 45° steps. |
| **Hand** | H | Drag to move your view of the canvas. |

For all the shape tools, hold ⇧ to keep the shape even (a square, a circle) and ⌥ to draw from the centre out.

## Colours and outlines

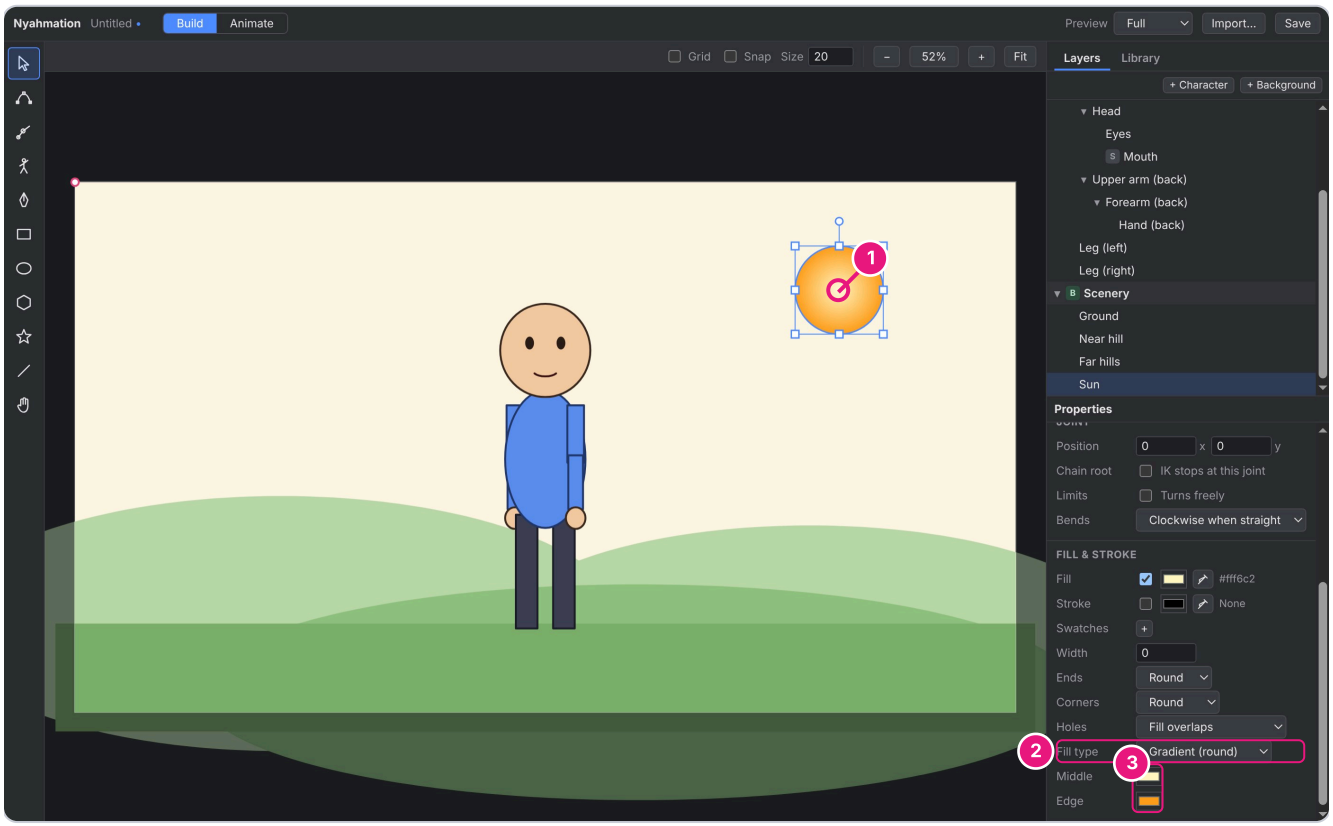
With a shape selected, the **Fill & stroke** section in Properties sets its colours. Untick **Fill** or **Stroke** to leave it out. The settings you choose are also used for the next shapes you draw. **Holes** decides what happens where a shape overlaps itself (for example, the middle of a letter O).

- **Eyedropper.** The pipette button beside each colour picks a colour from anywhere on the screen: another shape, a reference picture, even another app. Esc cancels.
- **Swatches** are the project's own palette, saved with it, so a character's skin and shirt colours are always one click away. Click a swatch to use it for the fill, ⇧-click for the outline, and ⌥-click to remove it. **+** adds the current fill colour.

## Gradients

Set **Fill type** to **Gradient (straight)** for colours that fade from one side to the other (a sunset, a shiny tube) or **Gradient (round)** for colours that fade from the middle outwards (a glowing sun, a ball). Pick

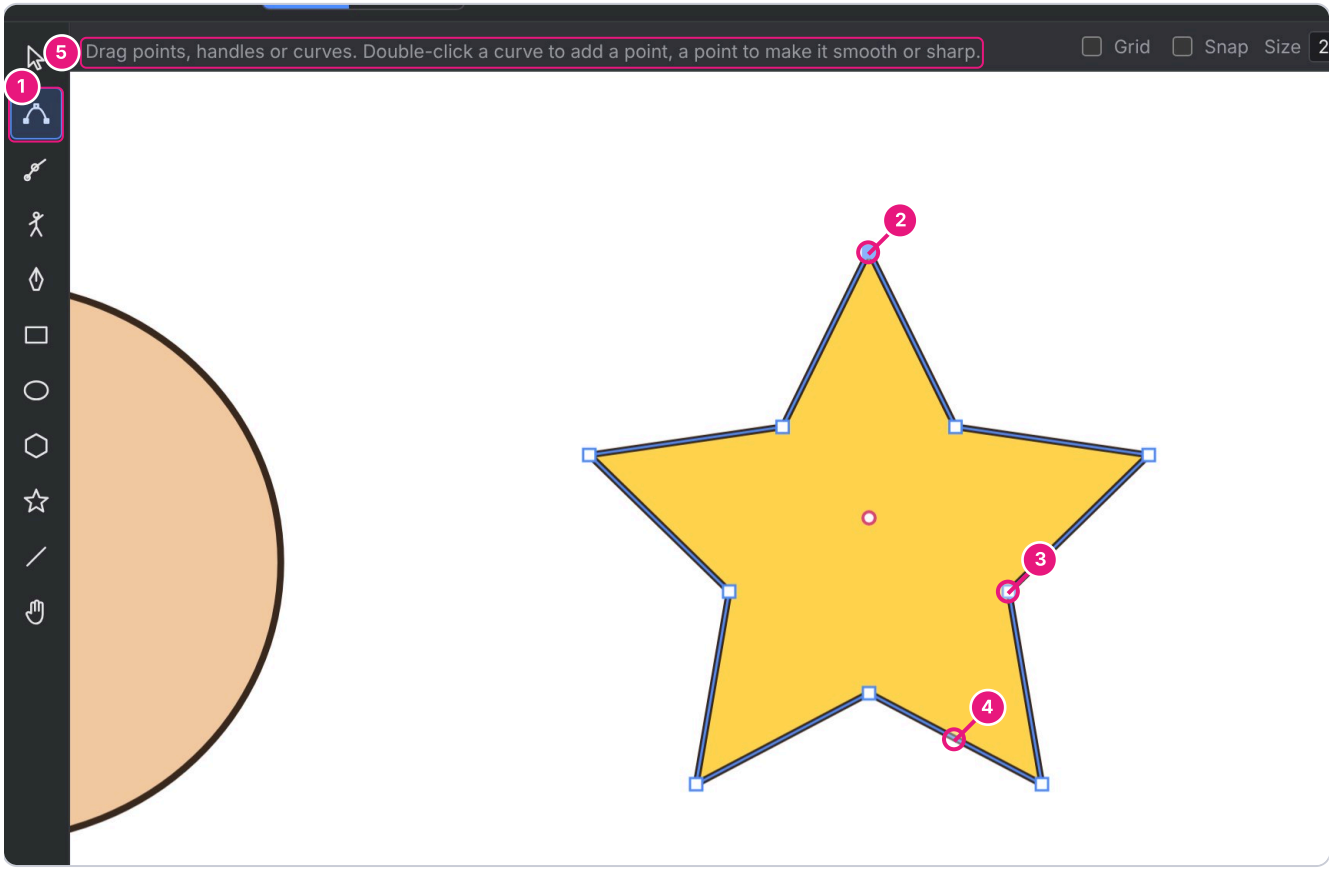
the two colours below it, and for a straight gradient its **Angle**. Gradients in imported SVG files are kept too.




*The demo's sun with a round gradient.*

1. The sun, now fading from pale yellow in the middle to orange at the edge
2. Fill type: Solid colour, Gradient (straight) or Gradient (round)
3. The gradient's two colours: the middle and the edge (for a straight gradient, the start and the end, plus an angle)

## Editing points




*Editing the star's points with the Points tool.*

1. Points tool (A)
2. A selected point. Drag to move it; Delete removes it
3. Another point of the shape. Double-click a point to make it smooth or sharp
4. Double-click anywhere on a curve to add a point
5. Hints for the current tool

- Drag a point, a curve, or a point's handles to reshape the shape.
- Double-click a curve to add a point; double-click a point to switch it between **smooth** (a round curve through it) and **sharp** (a corner).
- Hold ⌥ while dragging a handle to bend just one side of a smooth point.
- Delete removes the selected points. Esc stops editing.

## Arranging

- **In front or behind:** Object → Bring Forward / Send Backward (⌘ + ] / ⌘ + [), and Bring to Front / Send to Back with ⇧.
- **Group** (⌘ + G) keeps several parts together so they move as one. **Combine Shapes** (⌘ + 8) melts the selected shapes into a single shape. To merge, cut or overlap outlines instead, use the Shape Tools below.
- Nudge the selection with the arrow keys (add ⇧ for 10 pixels at a time).

## Shape tools: making new shapes from overlaps

Select two or more overlapping shapes in Build mode and choose one of the **Object → Shape Tools**. Think of cookie dough and cutters:

| TOOL | KEY | WHAT YOU GET |
|---|---|---|
| **Union** | ⌥ + ⌘ + U | One shape covering all of them, like pressing lumps of dough together. Good for a cloud made of circles. |
| **Subtract** | ⌥ + ⌘ + S | The back shape with the front shapes cut out of it, like a cookie cutter. Good for a crescent moon or a bite out of an apple. |
| **Intersect** | ⌥ + ⌘ + K | Only the part where they all overlap. Good for a lens or a leaf. |
| **Exclude** | ⌥ + ⌘ + E | Everything except where they overlap; the overlaps become holes. |

The result keeps the back shape's name, colours and effects, and its curves stay smooth and editable with the Points tool. It's one undo step.

## Cut, copy and paste

In Build mode, **Edit → Cut, Copy and Paste** (⌘ + X, ⌘ + C, ⌘ + V) work on the selected parts, or on a whole layer if its row is selected. Pasted parts land 20 pixels down and to the right of the last paste, inside the selected group or the active layer. You can paste into another project too: copy, open the other project, paste. Mouths bring their drawings with them. Animation isn't copied with parts; copy poses in Animate mode instead (Chapter 10).

> **Typing?** While you're typing in a text field, the same keys cut, copy and paste the text instead.

## Importing art

**File → Import Art or Sound...** (⌘ + I) brings in one file or several at once. Or simply **drag files from Finder onto the window**: pictures land where you drop them on the canvas, sounds go on the Sound row, and a **.nyah** project opens. Nyahmation brings in:

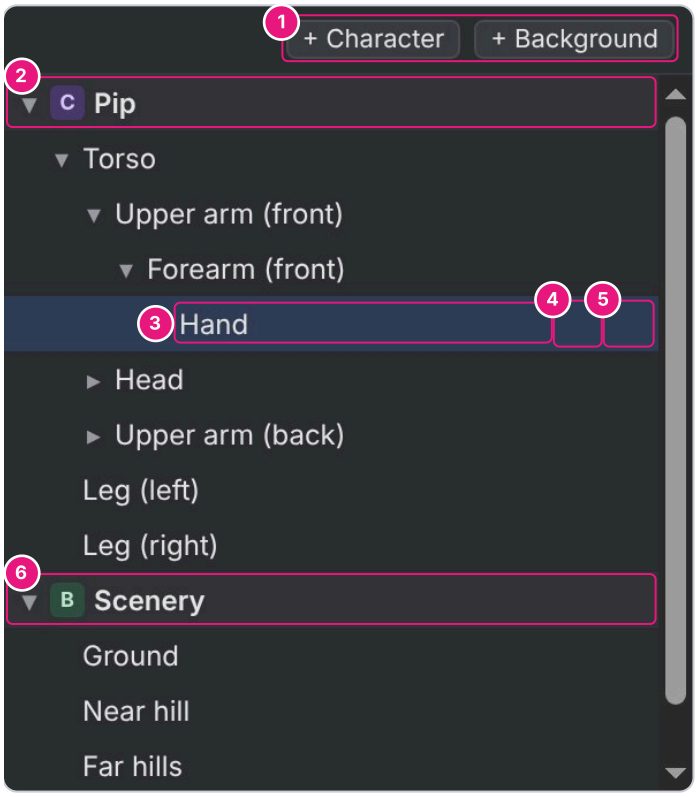
- **SVG** files from other drawing apps (Inkscape, Illustrator, Affinity, Figma...). Each layer or group in the file becomes a part, with its name, so a character drawn elsewhere arrives ready to rig. Gradients come along. Things Nyahmation can't use, such as text, are listed in a message.
- **PNG** and **JPEG** pictures. PNGs can have see-through backgrounds. Pictures have a fixed number of pixels, so they look soft if you blow them up; Nyahmation warns you when you export.
- **Sounds:** see Chapter 14.

CHAPTER 5

# Layers and parts

Characters, backgrounds and the part tree

A scene is a stack of **layers**. Each layer holds **parts**, and parts can hold other parts. The Layers panel shows all of it, front to back.




*The Layers panel with the demo puppet.*

1. Add a character layer or a background layer
2. A character layer (C) holds one rigged character
3. Parts inside parts: the hand is inside the forearm, inside the upper arm
4. Show or hide (appears when the mouse is over a row)
5. Lock, so it can't be selected or drawn on
6. A background layer (B) sits behind the characters

## Two kinds of layer

| LAYER | USE IT FOR |
|---|---|
| **Character (C)** | One character per layer. Its parts can be rigged and posed, and the whole character can be saved to the library. |
| **Background (B)** | Scenery: sky, hills, a room. It sits behind the characters (drag it up the list to put it in front, for things like a table the character stands behind). Backgrounds can have depth, scroll, and stay fixed to the camera (Chapter 12). |

## The part tree

A part inside another part moves with it, like a hand attached to a forearm attached to an upper arm. This is what makes a puppet: move the upper arm and everything below it comes along.

- **Put a part inside another:** drag its row onto the other part's row.
- **Reorder:** drag a row onto the top or bottom edge of another row.
- **Rename:** double-click a name. Good names help: they appear on the timeline, mouth shapes are recognised by their names, and Mirror Pose pairs parts by their names (Chapter 10).
- **Hide** with the eye, **lock** with the padlock. A locked part can't be selected, and nothing can be drawn on a locked layer.
- Select the parent of the selected part with ⇧ + Enter.
- **Copy a whole layer** (a character with all its parts) by selecting its row and pressing ⌘ + C, then ⌘ + V, here or in another project.

> **Tip.** Clicking on the canvas selects the exact part under the mouse (the hand, not the whole arm). Use the Layers panel, or ⇧ + Enter, to select something bigger.

## What's in front

Inside a character, which part is drawn in front is separate from the part tree. That means a back arm can be inside the torso (so it moves with it) and still be drawn behind it. Use Bring Forward and Send Backward to change it. In Animate mode the same commands change the order from the current frame on, for an arm that swings behind the body (Chapter 8).

CHAPTER 6

# Rigging a character

Joints, chain roots, limits and posing

Rigging means telling Nyahmation where each part bends: the shoulder, the elbow, the knee. It takes a few minutes and then posing is a matter of dragging.




*The Joints tool shows the skeleton of the demo puppet.*

1. Joints tool (J): shows the skeleton
2. A joint: where a part turns (here, the elbow). Drag it to move the joint without moving the drawing
3. Bones link each joint to the joints of the parts inside it
4. A chain root (a square): posing a hand stops here, so the body doesn't tip over
5. Joint settings: exact position, chain root, limits and bend direction

## Rigging a character, step by step

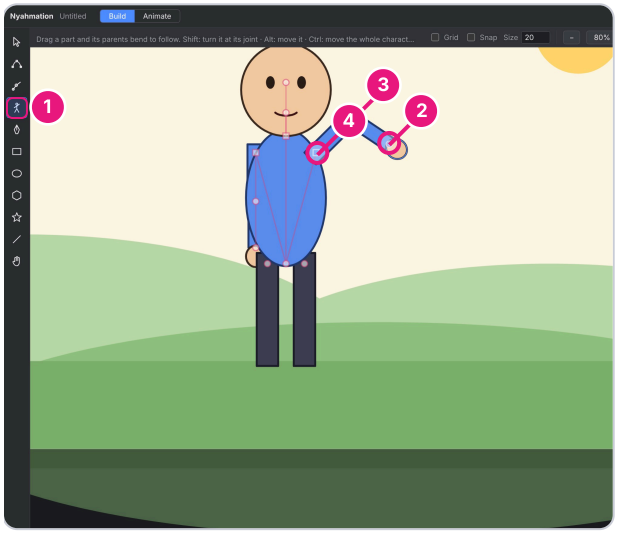
1. Put the character on a character layer, drawn as separate parts: head, torso, upper arm, forearm, hand, and so on.
2. In the Layers panel, build the tree: drag each hand onto its forearm, each forearm onto its upper arm, each upper arm onto the torso, and the head onto the torso.
3. Choose the Joints tool ( J ) and drag each part's joint to where it attaches: the shoulder, elbow, wrist, neck, hips and knees. The drawing stays put; only the turning point moves.

4. Select the character's layer and click **Mark branch joints as chain roots** in Properties (or use the Object menu). This stops a dragged hand from tipping the whole body over.
5. Try it with the Pose tool.

## Joint settings

| SETTING | WHAT IT DOES |
|---|---|
| **Position** | The exact position of the joint. |
| **Chain root** | Posing stops at this joint. Double-clicking a joint with the Joints tool does the same; chain roots are drawn as squares. |
| **Limits** | Stop a joint turning too far, so an elbow can't bend backwards. Set the smallest and largest angle. |
| **Bends** | Which way a straight arm or leg bends first when you pull on it (clockwise or counter-clockwise). |

## Posing with the Pose tool




*Dragging the hand upward: the elbow bends and the shoulder swings to follow.*

1. Pose tool (K)
2. Drag the hand: it follows the mouse...
3. ...the elbow bends...
4. ...and the shoulder swings. The joints that will turn are highlighted

With the Pose tool (K), drag any part and the parts it's attached to bend to follow, the way a real arm does. This is called **inverse kinematics** (IK). Hover over a part first to see which joints will turn. In Build mode, posing sets the character's resting pose; in Animate mode, it records a pose on the current frame.

| DRAG WITH... | WHAT HAPPENS |
|---|---|
| **Nothing held** | The parents bend to follow (the elbow and shoulder turn when you drag the hand). |

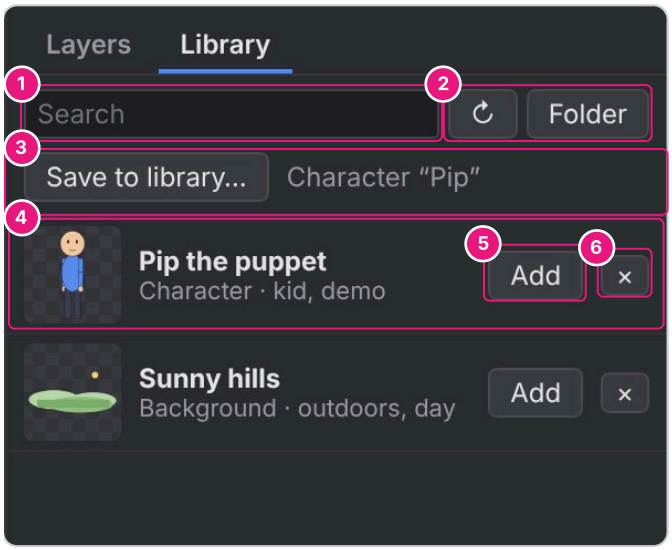
| DRAG WITH… | WHAT HAPPENS |
| --- | --- |
| ⇧ Shift | The part turns at its own joint only (just the hand turns at the wrist). |
| ⌥ Option | The part moves away from its joint. |
| ⌘ Command | The whole character moves. |

CHAPTER 7

# The library

Reusing characters, backgrounds and shapes

The library is your collection of reusable characters, backgrounds and shapes, shared by all your projects. It's a folder of files in **Documents/Nyahmation Library**.




1. Search by name or tag
2. Reload, and open the library folder in Finder
3. Save the selected layer or parts as a library item
4. A saved item, with a picture, its kind and tags
5. Add a fresh copy to this project
6. Move the item to the Trash

*The Library tab, with Pip and a background saved.*

## Saving to the library

1. Select a character's layer (click its row in the Layers panel), or some parts.
2. Open the **Library** tab and click **Save to library...**
3. Give it a name and some tags (words to find it by, separated by commas), and click **Save**. A picture is made automatically.

A background keeps its depth, scrolling and fixed-to-camera settings in the library (Chapter 12).

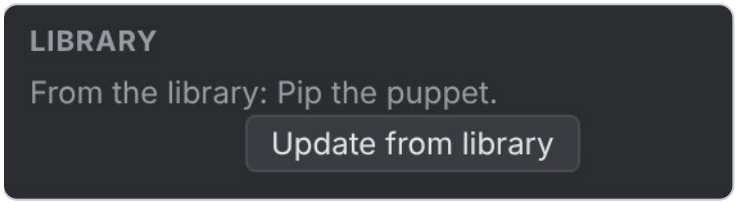
## Using a library item

Click **Add**. The project gets its own copy, so you can change it freely without changing the library, and you can add the same item as many times as you like (a crowd of Pips!). Shapes land where they were when you saved them.

## Updating a character everywhere

Say Pip gets a new hat. The projects that already use Pip each have their own copy, so they don't change by themselves; you decide when they do. Think of it like a recipe card: each project baked its own cake from the card, and updating the card doesn't change cakes already baked until you bake again.

1. Improve the character in any project, select its layer, and click **Save to library…**. Because the layer came from the library, the form offers **Save as a new version of "Pip the puppet" (replaces it)**, already ticked. Click **Save**.
2. In another project that uses Pip, select Pip's layer. Properties show where it came from, and say when the library has a newer version.
3. Click **Update from library**. The new drawings and rig come in, and **the animation is kept**: every part that still exists keeps its poses. The status bar says how many parts kept their animation, and how many are new or gone.




*A layer that came from the library, in Layer properties.*

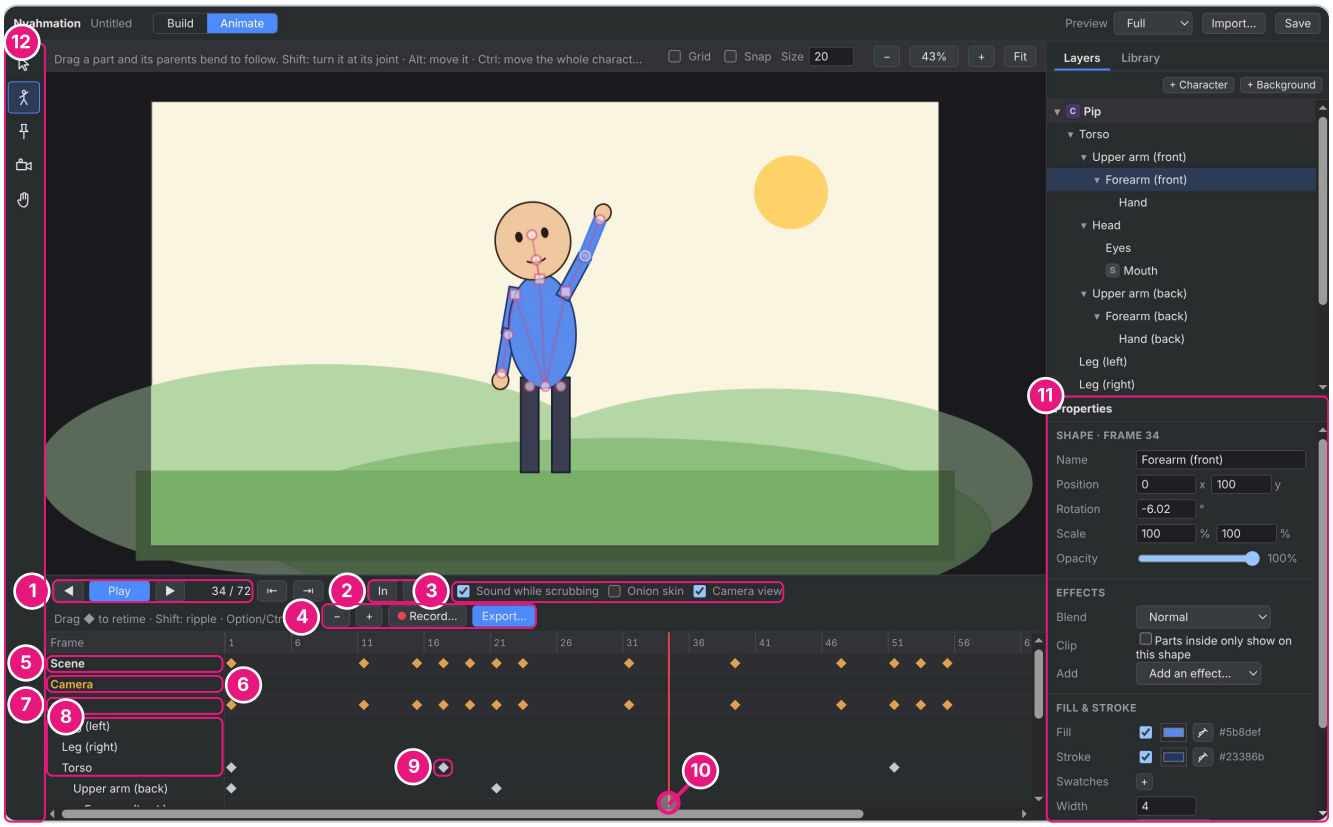
> **Organising.** Click **Folder** to open the library in Finder. Make subfolders there, such as **Mouths** or **Characters/Animals**, and move items into them; the Library tab shows the folders. Click ↻ after making changes in Finder.

CHAPTER 8

# Animating

Poses, the timeline and playback

Switch to **Animate** mode with the button at the top. The timeline appears at the bottom of the window.




*Animate mode: the timeline at the bottom, with a row for every part.*

1. Play and pause (Space), step a frame (← →), and the current frame
2. Loop range: I and O set where playback loops
3. Sound while scrubbing, onion skin, and camera view
4. Timeline zoom, and Export
5. The Scene row: every pose in the scene
6. The Camera row: the camera's moves (Chapter 11)
7. A layer row: every pose of that character
8. Part rows: each part's own poses
9. A pose (◆). Drag it to retime
10. The playhead: the frame on screen
11. Properties show and record the pose on this frame
12. Animate mode tools: Select, Pose, Pin, Camera and Hand

## Making your first animation

1. Open the demo puppet, or your own rigged character, and switch to **Animate**.
2. Click frame 12 on the timeline's number row (or press <kbd>→</kbd> eleven times).

3. Choose the Pose tool ( K ) and drag the hand up. A ◆ appears on the rows of the parts that moved, on frame 12.
4. The first time a part moves, its starting position is also kept on frame 1 automatically, so the arm moves *from* where it was.
5. Press Space to play. The arm rises smoothly from frame 1 to frame 12.
6. Go to frame 24 and pose the hand down again. Now it waves up and down.

## How poses work

- Only the parts you change get a pose. The rest keep doing whatever they were doing.
- Between two poses, the motion is smooth: it speeds up and slows down naturally and flows through poses instead of stopping at each one.
- You can also pose by typing numbers in Properties (position, rotation, scale, opacity, and effect settings), or with the Select tool. Any change on a frame becomes a pose.

## The timeline rows

| ROW | WHAT ITS ◆ MARKS ARE |
|---|---|
| **Sound** | Your sound clips, with their waveforms (Chapter 14). |
| **Scene** | Every frame that has a pose on it anywhere, camera included. Dragging a mark here retimes everything on that frame. |
| **Camera** | The camera's moves (Chapter 11). |
| **Layer (Pip)** | Every pose of one character. Click the triangle to show or hide its parts. Changes of Animate on, and cycles, show here too (Chapter 9, Chapter 10). |
| **Part rows** | Just that part's poses, including draw-order changes and effect settings. |

## Playing it back

- Space plays and pauses. ← → step one frame; ⇧ + ← ⇧ + → jump to the previous or next pose. Home and End go to the start and end (on a MacBook keyboard: Fn + ← and Fn + →).
- **Loop range:** press I on the first frame and O on the last. Playback repeats just that part, which is handy while you polish one move. × clears it.
- Drag the top edge of the timeline to make it taller, and use **−** / **+** to zoom it.

## Smooth playback on a slower computer

While a scene plays, the timeline bar shows **Showing 24 of 24 fps**: how many frames a second your computer is actually drawing. If it turns orange (say, **Showing 15 of 24 fps**), some frames are being skipped to keep up with the sound. The picture stays in time; it's just less smooth. Effects such as blur are the usual cause.

- **Preview** in the top bar draws the canvas at **Full**, **Half** or **Quarter** resolution. Lower is softer but much faster, like squinting at the screen. Handles and outlines stay sharp. Nyahmation remembers the choice on this computer.
- **View → Measure Preview Speed** times your scene at each quality and recommends one.

> **Exports are never affected.** The Preview setting only changes what you see while working. Exported videos always draw every frame at full quality, however long that takes.

## Changing what's in front, over time

An arm that swings behind the body needs to change its place in the stacking order partway through the move. In Animate mode, select the part, go to the frame where it should pass behind, and choose **Send Backward** (⌘ + [) or **Send to Back** (⇧ + ⌘ + [); **Bring Forward** and **Bring to Front** go the other way. A mark on the part's row records the change, and the new order holds until the next change. Drag the mark to retime it like any other pose.

## Scene settings

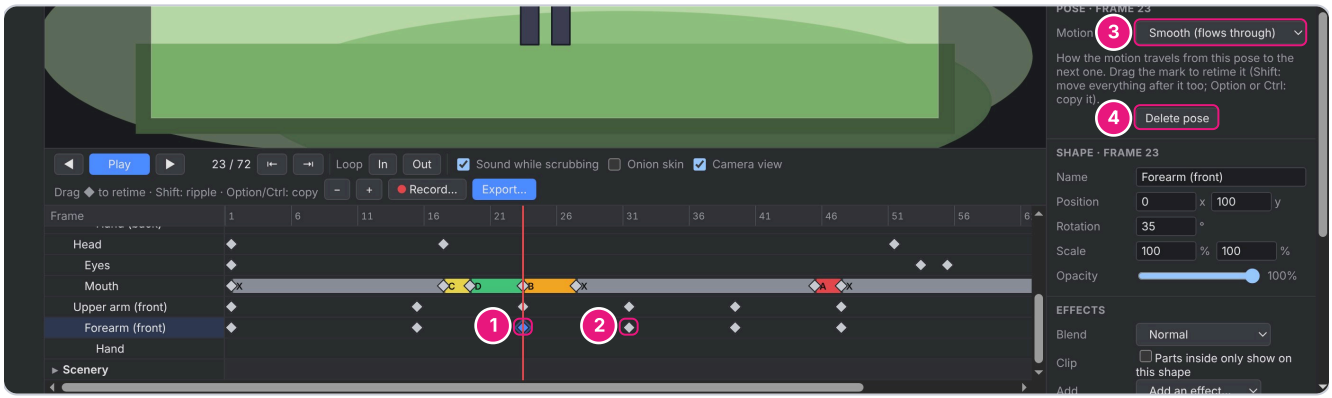
With nothing selected (press Esc), Properties shows the scene: its size in pixels (1920 × 1080 is standard HD video), the frame rate (24, 25, 30 or 60 frames a second; 24 is the classic cartoon rate; changing it keeps everything's timing in seconds, so moves keep their speed and lip sync stays on the words), the length in frames, the background colour, an optional **Sky** gradient (Chapter 12), and **Animate on** (Chapter 9).

CHAPTER 9

# Timing

Retiming, easing, stretching, on twos, onion skin and pins

## Retiming: making moves faster or slower




*A selected pose, and the easing choices for it in Properties.*

1. The selected pose (blue)
2. Drag a pose left to make the move faster, right to make it slower. Shift-drag moves everything after it too
3. How the motion leaves this pose: Smooth, Ease in/out, Linear, Hold or a custom curve
4. Delete the selected poses (or press Delete)

Timing is what makes animation feel alive, and in Nyahmation it's just dragging:

- **Drag a ◆** to another frame. A pose on frame 10 dragged to frame 5 makes that move twice as fast.
- ⇧**-drag** moves the pose *and everything after it* on that row, so the timing of the rest of the scene is kept.
- ⌥**-drag** (or Ctrl-drag) **copies** the pose to another frame. Two copies of the same pose make a **hold**: the character stays still between them.
- Click a ◆ to select it (⇧-click for more), then press **Delete** to remove it.
- Which row you drag on decides what moves: the Scene row moves every part's pose on that frame (and the camera's), a layer row moves one character's, and a part row moves just that part's.

> **Lip sync stays put.** Dragging on the Scene or a layer row never moves the mouth shapes, because the voice they match doesn't move. To shift the mouth, drag on the Mouth's own row.

## Stretching a whole stretch of animation

Dragging poses one by one is fine for one move. To make a whole passage slower or faster, set a loop range around it (**I** and **O**), then type a new length into **Stretch to … frames** on the timeline bar. The poses inside spread out or squeeze together evenly, like stretching a spring, and every pose after the range moves along so the rest keeps its timing.

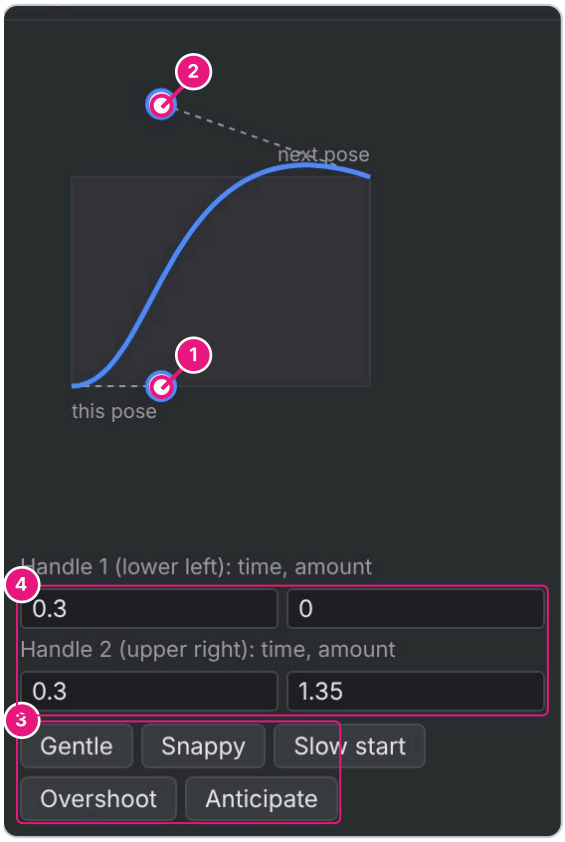
- With parts selected, only their animation is stretched (and the parts inside them). With nothing selected, the whole scene and the camera are.
- Lip sync isn't stretched unless you select the mouth, because the dialogue doesn't change speed.

## Easing: how a move starts and stops

Select a pose and choose **Motion out** in Properties. It decides how the move from this pose to the next one travels:

| CHOICE | LOOKS LIKE |
|---|---|
| **Smooth** | The default. Flows naturally through the poses. |
| **Ease in and out** | Starts slowly, speeds up, and settles gently. Good for a move that starts and stops. |
| **Ease in** | Starts slowly, ends fast. Good for a fall or a throw. |
| **Ease out** | Starts fast, ends slowly. Good for landing or coming to rest. |
| **Linear** | The same speed all the way. Mechanical things. |
| **Hold** | Doesn't move at all, then jumps to the next pose. Snappy, cartoony changes. |
| **Custom curve…** | Your own timing, drawn as a curve (below). |

### Drawing your own timing curve




1. Handle 1: drag it to shape how the move leaves this pose
2. Handle 2: how the move arrives at the next pose. Above the top, it overshoots and settles back
3. Presets to start from
4. The handles' exact numbers

*A custom curve with a little overshoot.*

The curve is a graph of the move: time runs left to right, and how far the part has got runs bottom to top. Steep means fast; flat means slow. Drag the two handles to bend it. Push a handle above the top for an **overshoot** (the arm swings a little past its pose and settles back) or below the bottom for an **anticipation** (a little wind-up the other way first, like crouching before a jump). The presets **Gentle**, **Snappy**, **Slow start**, **Overshoot** and **Anticipate** are good starting points.

## On ones, twos and threes

Classic hand-drawn cartoons often show each drawing for two frames ("on twos"), which gives a lively, drawn feel. Set **Animate on** in the scene's Properties to **Ones**, **Twos** or **Threes**, or change it for just one character in its layer's Properties. Mouths always stay on ones, so lip sync stays accurate, and so does the camera.

- **Changing it partway through.** Fast action often looks best on ones, and the calm bits on twos. In Animate mode, select a character's layer, go to the frame where the change should happen, and pick **Animate on** in its Properties. A mark on the layer's row records it, and it lasts until the next change.
- **View on ones.** When anything is on twos or threes, a **View on ones** checkbox appears on the timeline bar. Tick it to see every frame smoothly while you work on the motion; the export still uses twos and threes.

## Onion skin




*Onion skin shows faint copies of the frames before and after this one.*

1. Onion skin is on
2. Earlier frames show in red…
3. …and later frames in green, so you can see the arc of a movement
4. How many frames each way, and how far apart they are

Tick **Onion skin** on the timeline to see faint copies of nearby frames: red for earlier, green for later. It shows the path a movement takes, so you can check that an arm swings in a nice arc. Next to it, choose how many frames to show each way (1 to 4) and how far apart they are (every frame up to every 4th).

## Pins: feet that stay on the floor




*The left leg pinned from frame 6.*

1. Pin tool (P)
2. The pinned foot. It stays put while the body moves; the leg bends to reach it
3. The yellow bar shows the frames where the part is pinned
4. The status bar confirms what happened

When a character walks or crouches, its feet should stay planted instead of sliding. That's what pins are for.

1. Go to the frame where the foot touches the ground.
2. Choose the Pin tool (`P`) and click the foot. It's pinned from this frame on.
3. Move the body in later frames: the foot stays exactly where it is, and the leg bends to reach it.
4. On the frame where the foot should lift, click it again with the Pin tool to release it.

The yellow bar on the part's row shows the frames where it's pinned. Pins hold the foot to the *stage*, so the camera can move all it likes without the foot sliding.

CHAPTER 10

# Walks and repeats

Copying poses, mirroring, and animation cycles

A walk is the classic animation exercise, and three tools make it much quicker: copying poses, mirroring them, and repeating a stretch of animation as a cycle. They're useful well beyond walks, too: a blink every few seconds, a flag flapping, a dog wagging its tail.

## Copying and pasting poses

In Animate mode, **Copy** (⌘ + C) copies the *pose* of the selected parts on this frame (select the character's layer row to copy the whole character). Then:

- **Paste on another frame** (⌘ + V) to repeat the pose later, for a hold or for the end of a cycle.
- **Paste onto another character** by selecting it first. Parts are matched by name, so this works between copies of the same character (or any character with the same part names).

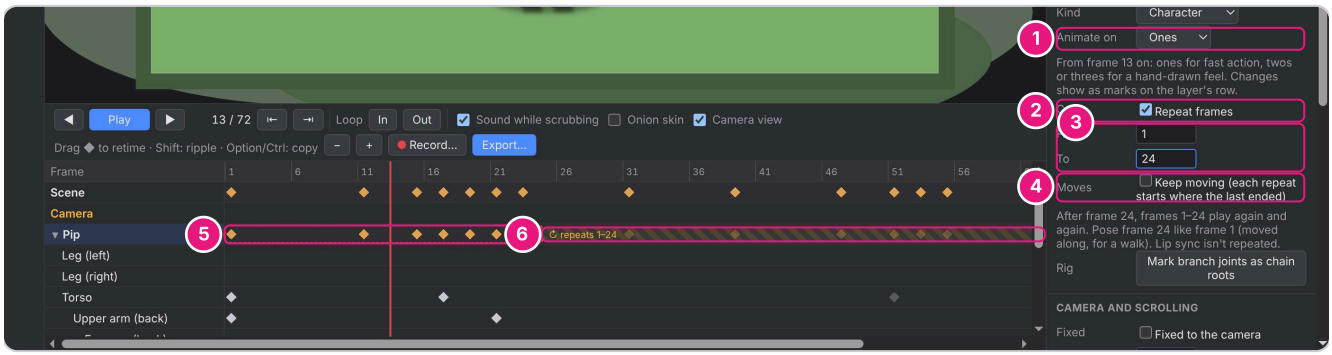
Mouth shapes aren't included, because they belong to the dialogue.

## Mirror Pose and Swap Sides

Both work in Animate mode on the selected parts, or on the whole active character if nothing is selected, and record poses on the current frame. They need to know which parts are a pair, and they find out from the names: **left**/**right**, **front**/**back**, or a lone **L**/**R** ("Arm L", "leg_R"). Parts inside a pair pair up too, so "Hand" inside "Forearm (front)" pairs with "Hand (back)" inside "Forearm (back)".

| COMMAND | KEY | WHAT IT DOES |
|---|---|---|
| **Mirror Pose** | ⌥ + ⌘ + M | Flips the pose side to side, like looking in a mirror: the left arm takes the right arm's pose turned the other way, and parts without a partner (the head, the torso) turn the other way themselves. For characters facing the camera: wave with the other hand in one keystroke. |
| **Swap Sides** | ⌥ + ⇧ + ⌘ + M | Paired parts swap poses exactly as they are; everything else stays. For characters seen from the side, this is the second half of a walk: the left leg does what the right leg just did. |

## Cycles: repeating animation




*Pip's layer repeating frames 1–24 for the rest of the scene.*

1. Animate on: ones, twos or threes from this frame on (Chapter 9)
2. Cycle: Repeat frames
3. The frames to repeat: From and To
4. Keep moving: each repeat starts where the last one ended
5. The cycle, outlined on the layer's row
6. "↻ repeats": the rest of the scene plays the cycle again and again. Poses after the cycle are dimmed, because they don't play

A cycle is like a record's groove: the needle goes round the same loop again and again. Select a character's (or background's) layer, tick **Repeat frames** in its Properties, and set **From** and **To**. If you've set a loop range, the cycle starts with it. After frame *To*, the layer plays frames *From* to *To* again, for the rest of the scene.

- **Frame To counts as frame From of the next repeat,** so pose them the same (copy and paste the pose). Otherwise the character jumps at every repeat.
- **Keep moving** makes each repeat start where the last one ended. The character's top part and anything pinned move on by the distance one cycle covers, so a walk actually walks, with its feet still planted.
- Lip sync is never repeated: mouth shapes follow the dialogue. Everything else in the layer repeats, including blinks, draw-order changes, pins and changes of Animate on.
- After frame *To* the layer only replays the cycle, so posing it there wouldn't show. Nyahmation doesn't record such a pose: the status line says which frame of the cycle is showing. Change the cycle itself, inside From–To.

## A walk in six steps

1. On frame 1, pose the first **contact**: the front foot's heel just touching the floor, the back foot's toe about to leave it, arms swinging opposite to the legs.
2. Copy the pose (⌘ + C), go to frame 13, paste (⌘ + V) and choose **Swap Sides** (⌥ + ⇧ + ⌘ + M). That's the second contact, with the other leg forward.
3. Go to frame 25 and paste again: the first contact once more, so the cycle closes.
4. Add the in-between poses: the **passing** positions on frames 7 and 19, with the body a little higher.
5. Pin the planted foot on each contact frame and release it when it lifts (Chapter 9).

6. Select the layer, tick **Repeat frames** from 1 to 25, and tick **Keep moving**. Pip now walks across the whole scene.

> **Walking on the spot.** For a character who stays in the middle while the world moves past, leave Keep moving off and scroll the background instead (Chapter 12). It's the treadmill trick of old cartoons.

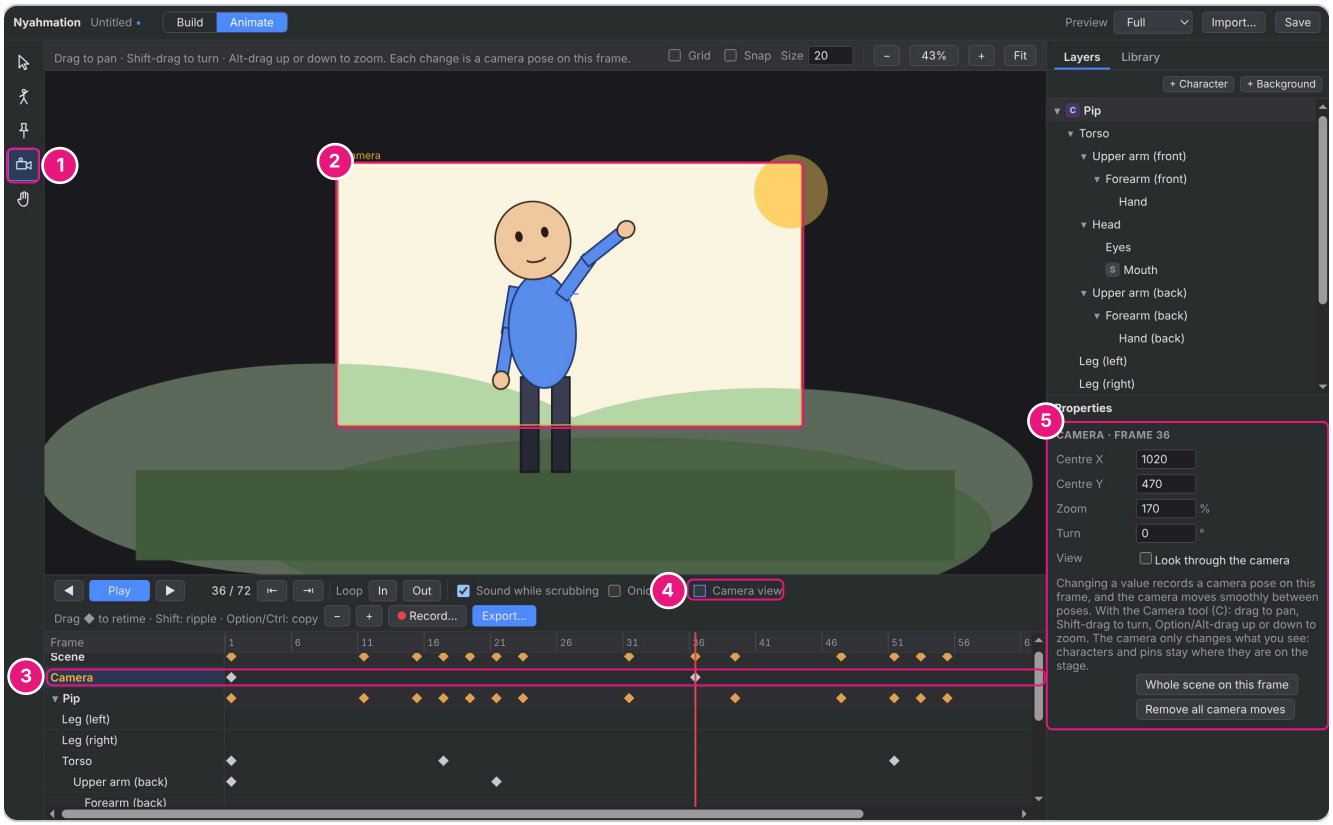
CHAPTER 11

# The camera

Panning, zooming and turning the view

Every scene has a camera. At first it simply looks at the whole scene, but you can move it on the timeline like any part: slowly push in on a face, pan across a landscape, or tilt the picture for a dizzy moment.

Imagine a real film set. The actors stand on the stage, and the camera operator moves the camera around them. The camera never pushes an actor. It's the same in Nyahmation: **the camera only changes what you see**. Characters, poses and pins stay exactly where they are on the stage, so a pinned foot never slides because the camera moved.




*Camera view is off, so the whole stage shows, with the camera's frame on it.*

1. Camera tool (C), in Animate mode
2. The camera's frame: what the video will show on this frame
3. The Camera row: a ◆ for each camera move
4. Camera view: on, you look through the camera; off, you see the stage with the camera's frame
5. Camera properties: exact position, zoom and turn

## Moving the camera

1. In Animate mode, choose the Camera tool ( C ) and go to the frame where the move should end.

2. Drag on the canvas to pan. The picture follows the mouse, as if you were sliding the stage under the camera.
3. ⌥-drag up or down to zoom in or out, and ⇧-drag to turn the picture around its middle.
4. Play it. The camera glides there from where it was, because its first move also keeps its starting place on frame 1.

Camera moves are poses like any other: they appear on the Camera row, can be dragged to retime them, and use the same easing (an **Ease in and out** camera move feels like a smooth dolly). Dragging on the Scene row retimes them together with everything else.

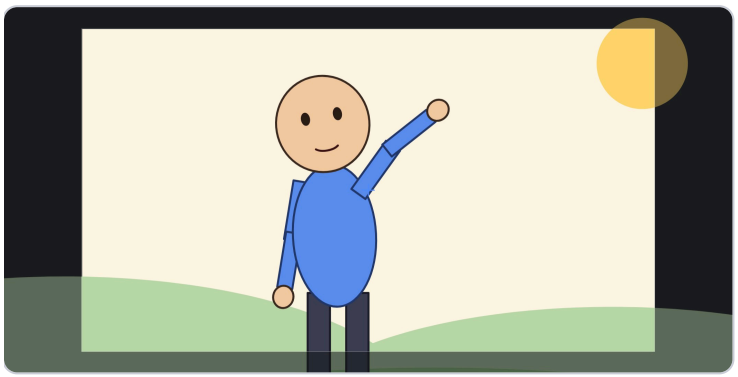
## Camera properties

With the Camera tool chosen, or the Camera row selected, Properties shows the camera on this frame:

| SETTING | WHAT IT DOES |
|---|---|
| **Centre X / Y** | The stage point in the middle of the picture. |
| **Zoom** | 100% shows the whole scene; 200% shows half as much, twice as big. |
| **Turn** | Tilts the picture, in degrees. |
| **View** | **Look through the camera**: the same as Camera view on the timeline. |
| **Whole scene on this frame** | A camera pose back at the start: the whole scene, straight, at 100%. |
| **Remove all camera moves** | Deletes every camera pose (you can undo it). |

## Camera view on or off

**Camera view** (on the timeline bar, on by default) shows the picture exactly as the video will. Turn it off to see the whole stage, with the camera's frame drawn on it in orange. That's handy when a character is about to walk into the shot and you need to pose it while it's still out of the picture. All the tools work either way.



*The frame above with Camera view on: what the video shows.*

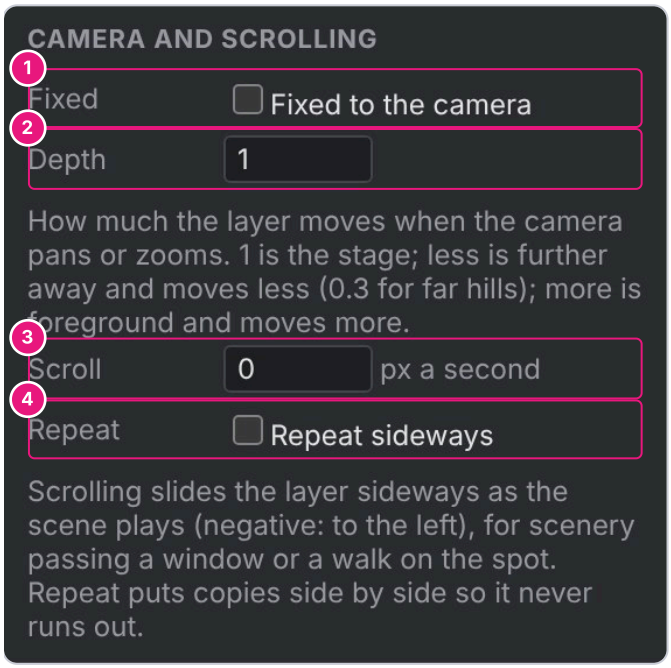
**Pictures and zoom.** Zooming in enlarges everything, including PNG pictures. The export window takes the camera's closest zoom into account when it warns about pictures that may look soft.

CHAPTER 12

# Backgrounds and depth

Skies, parallax, fixed layers, followers and scrolling

Once there's a camera, backgrounds can do much more than sit there. Each layer's Properties have a **Camera and scrolling** section.




*Camera and scrolling, in a layer's Properties.*

1. Fixed to the camera: stays put on screen
2. Depth: how far away the layer is
3. Scroll: slides the layer sideways as the scene plays
4. Repeat sideways: copies side by side, so it never runs out

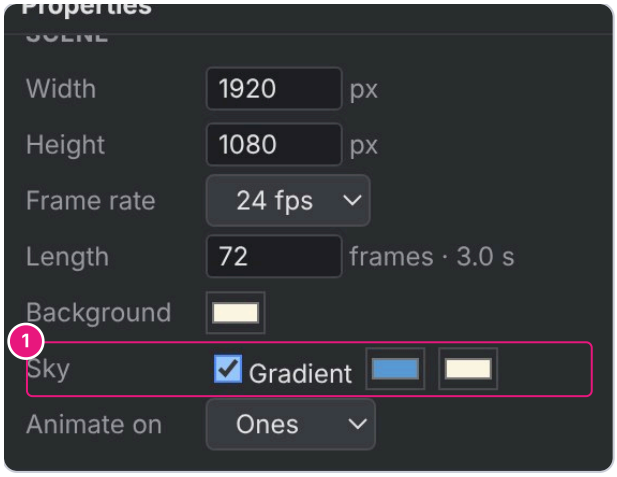
## Depth: near things move more

Look out of a train window: the fence next to the track whizzes past, the trees go by more slowly, and the mountains hardly move. That difference, called **parallax**, is how our eyes judge distance, and it makes a flat cartoon feel deep.

Give each background layer a **Depth**. **1** is the stage, where the characters are. Less than 1 is further away and moves less when the camera pans or zooms: try 0.3 for far hills and 0.6 for nearer ones. More than 1 is foreground and moves more: 1.5 for a bush right in front of the camera. Depth only shows when the camera moves, so add a camera pan to see it.

## A sky behind everything

Press Esc to see the scene's Properties and tick **Sky: Gradient**. Pick the top and bottom colours. The sky fills the whole picture, fading from top to bottom, and stays put whatever the camera does, like a real sky.



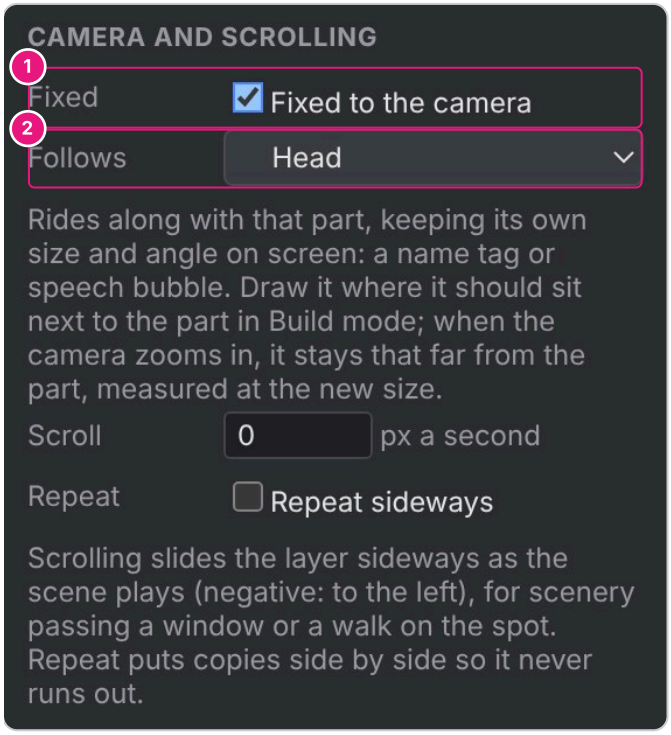

*The scene's Sky setting: a gradient from the top colour to the bottom colour.*

1. Sky: tick Gradient, then pick the top and bottom colours

## Fixed to the camera: titles and corners

Tick **Fixed to the camera** for a layer that should stay in the same place on screen, at the same size, whatever the camera does: a title, a logo, a narrator in the corner. It's like a sticker on the camera's lens.

## Following a part: tags and speech bubbles




*A speech bubble layer that follows Pip's head.*

1. Fixed to the camera, ticked
2. Follows: the part to ride along with

A fixed layer can also **follow** a part in another layer. It rides along with the part but keeps its own size and angle, so a name tag stays readable and a speech bubble stays level when the head tilts or the camera zooms in.

1. Make a new background layer (**+ Background**) and, in Build mode, draw the bubble or tag where it should sit next to the character.
2. Select the layer, tick **Fixed to the camera**, and choose the part in **Follows** (the head, say).

When the camera zooms in, the bubble stays the same size on screen, and its distance from the head grows with the head, so it never ends up on top of the face.

## Scrolling scenery

**Scroll** slides a layer sideways as the scene plays, in pixels a second; negative numbers go left. Tick **Repeat sideways** and copies of the layer are laid side by side, so it never runs out. Use it for clouds drifting by, scenery passing a car window, or the ground under a character walking on the spot.

> **Treadmill walk.** Put the character in the middle with a walk cycle (without Keep moving), and scroll the ground and scenery at walking speed, the far layers more slowly than the near ones.

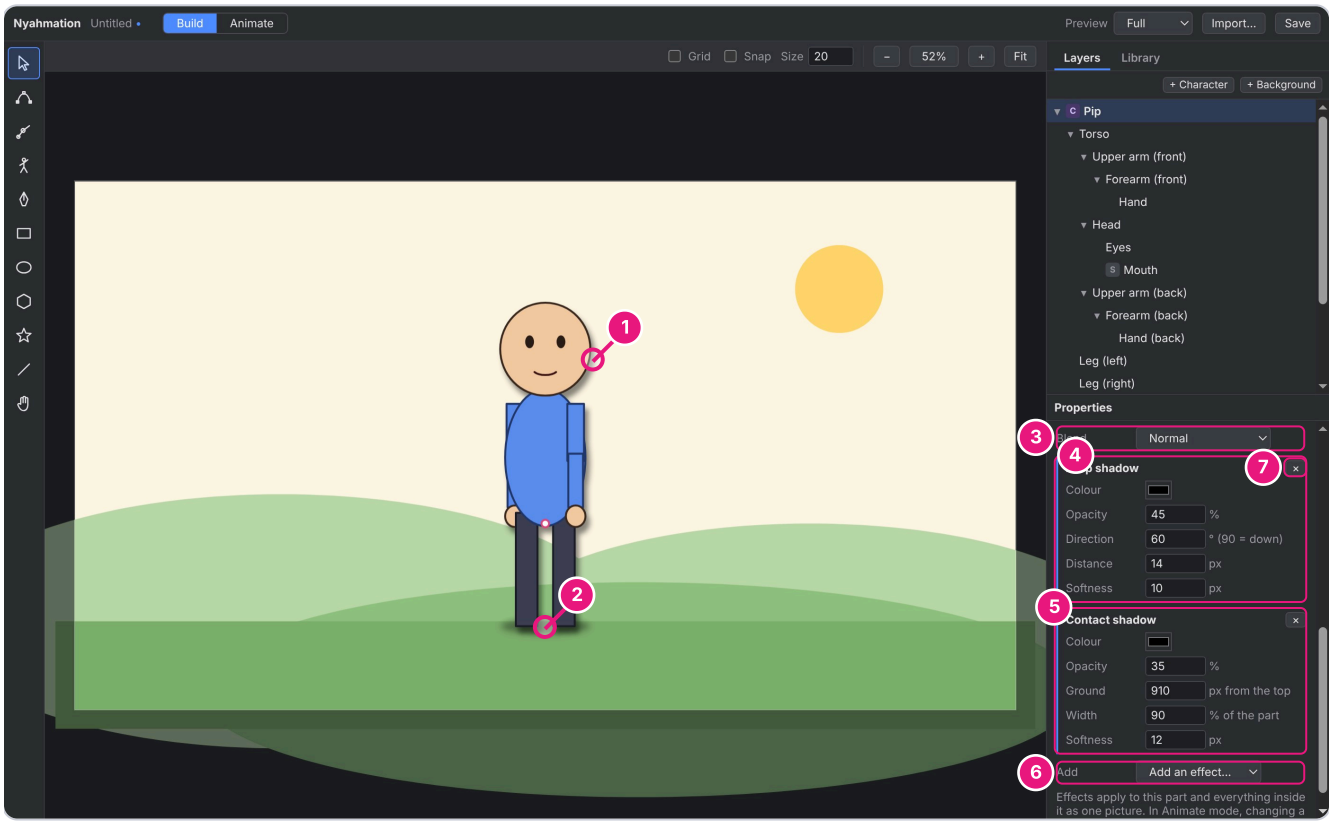
Background layers saved to the library keep their depth, scrolling and fixed setting. (A "follows" link isn't kept, because the part it named belongs to another project.)

CHAPTER 13

# Effects

Shadows, glows, blur, haze, blend modes and clipping

Effects add the finishing touches: a shadow to lift a character off the background, a glow for magic, haze to push far hills into the distance.




*Pip's layer with a drop shadow and a contact shadow.*

1. A drop shadow of the whole character
2. A contact shadow on the ground under his feet
3. Blend: how the layer mixes with what's behind it
4. The drop shadow's settings
5. The contact shadow's settings
6. Add an effect...
7. Remove an effect (and its animation)

## Adding an effect

Select a part (its Properties have an **Effects** section) or a layer's row (**Layer effects**, for the whole layer), and choose from **Add an effect...**. × removes one.

| EFFECT | SETTINGS | GOOD FOR |
|---|---|---|
| **Drop shadow** | Colour, opacity, direction, distance, softness | Lifting a character or a title off the background. |

| EFFECT | SETTINGS | GOOD FOR |
|---|---|---|
| **Outer glow** | Colour, opacity, size, strength (1 is normal, up to 4) | Magic, lamps, a glowing sun, neon signs. |
| **Blur** | Amount | Things far away or out of focus, and speed. |
| **Haze** | Colour, amount | Fading distant layers toward the sky colour, as air does with real mountains. |
| **Contact shadow** | Opacity, ground (how far down the stage the floor is; it starts at the part's lowest point), width (a share of the part), softness | A soft oval on the ground under a character. As the character jumps, it shrinks and fades. |

## Effects apply to the whole group

A shadow on a character's layer casts *one* shadow of the whole character, the way a real shadow works. If every part cast its own, the shadows would double up and look darker wherever parts overlap, such as the arm over the body. In the same way, an effect on a group (an arm with its forearm and hand) applies to the group as one picture.

## Animating effects

In Animate mode, changing an effect's number records a pose, like moving a part: a glow can pulse, a shadow can lengthen as the sun sets, a blur can sharpen as something comes into focus. The marks appear on the part's row and retime like any other. Colours don't animate.

## Blend modes

| BLEND | DOES |
|---|---|
| **Normal** | Covers what's behind. |
| **Multiply (shading)** | Darkens what's behind, like a coloured filter. Draw shading shapes in a soft blue or purple on Multiply. |
| **Screen (light)** | Lightens what's behind: highlights and light beams. |
| **Add (bright light)** | Lightens even more strongly: sparks, fire, glows. |
| **Overlay (contrast)** | Makes lights lighter and darks darker. |

## Clipping: pupils that stay in the eye

Tick **Clip** (“Parts inside only show on this shape”) on a shape, and the parts inside it only show where the shape itself is. Put the pupil inside the eye white and clip: the pupil can look all the way to the side

and never pokes out of the eye. It works for shading inside a body, a pattern on a shirt, or a face in a window.

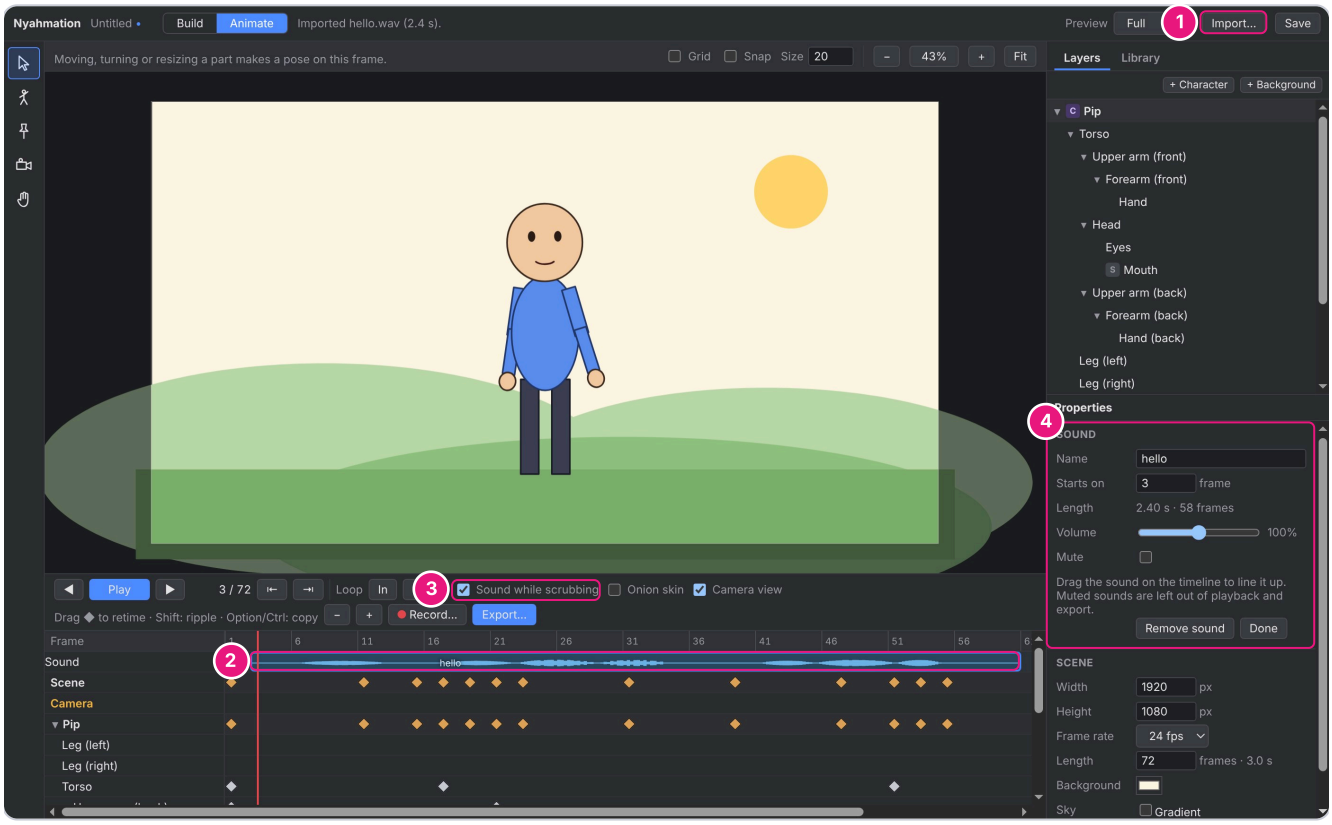
> **Speed.** Blur and big glows are the slowest things Nyahmation draws. Parts that don't change from one frame to the next (most scenery) are drawn once and reused, so they cost little. If playback stutters, lower the Preview quality (Chapter 8); exports are always full quality. Onion skins leave effects out, to stay quick.

CHAPTER 14

# Sound

Adding dialogue and music

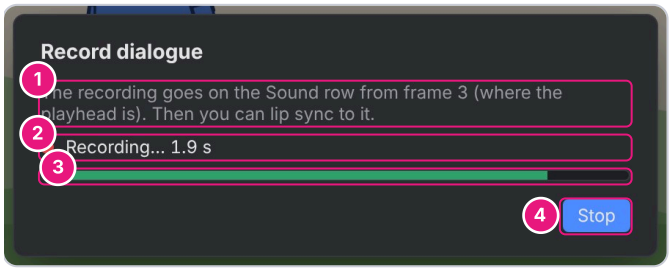
Add a voice recording, sound effects or music. Nyahmation plays it in time with the picture and puts it in the exported video.




*A voice recording on the Sound row, selected, with its properties.*

1. Import... (⌘I) adds a sound file at the current frame; Record... on the timeline records one
2. The Sound row, with the sound's waveform. Drag it to line it up
3. Sound while scrubbing: hear each frame as you step or drag the playhead
4. Sound properties: name, start frame, volume and mute

## Recording dialogue




*Recording a line of dialogue.*

1. Where the recording will start: the playhead
2. Recording, and for how long
3. How loud the microphone hears you
4. Stop when you've finished the line

1. In Animate mode, go to the frame where the line should start.
2. Click **Record...** on the timeline (or **File → Record Dialogue...**, ⇧ + ⌘ + R), then **Start recording**, and say the line. The bar shows how loud the microphone hears you.
3. Click **Stop**. Listen to it, then click **Keep it**, or **Try again**. A kept recording goes on the Sound row, named Line 1, Line 2 and so on, ready for lip sync.

The first time, macOS asks whether Nyahmation may use the microphone: click **Allow**. If you clicked Don't Allow, turn it on in **System Settings → Privacy & Security → Microphone**. Use headphones if the scene already has sounds, so the microphone doesn't hear them.

## Adding a sound file

1. In Animate mode, go to the frame where the sound should start.
2. Choose **File → Import Art or Sound...** (⌘ + I) and pick a sound file: WAV, MP3, M4A, AAC, OGG or FLAC.
3. It appears on the Sound row with its **waveform** (the wiggly shape of the sound: tall where it's loud, flat where it's quiet). If the sound is longer than the scene, the scene gets longer to fit it.

## Working with sounds

- Move a sound by dragging it along the Sound row.
- Click it to see its properties: name, start frame, volume (up to 200%) and mute. Press Delete to remove it.
- **Sound while scrubbing** (a checkbox on the timeline, on by default) plays a tiny slice of sound for each frame as you step with the arrow keys or drag along the number row. This is how you find exactly where each word starts.
- During playback the picture follows the sound, so they never drift apart, even on a slow computer.

> **Recording tip.** Record in a quiet room, with the microphone close. Speak a little more clearly and slowly than normal; it makes lip sync easier, for you and for Auto lip sync. A recording from your phone's voice memo app works too: share it to the Mac and drag it onto the window.

CHAPTER 15

# Mouths and lip sync

Making a mouth and syncing it to speech

A talking character swaps between a handful of mouth drawings, one for each group of sounds. In Nyahmation the mouth is a **switch layer**: a part that shows one drawing at a time. You can type the letter of each mouth shape as you hear the dialogue, or let Nyahmation listen and fill them in for you.

## The nine mouth shapes

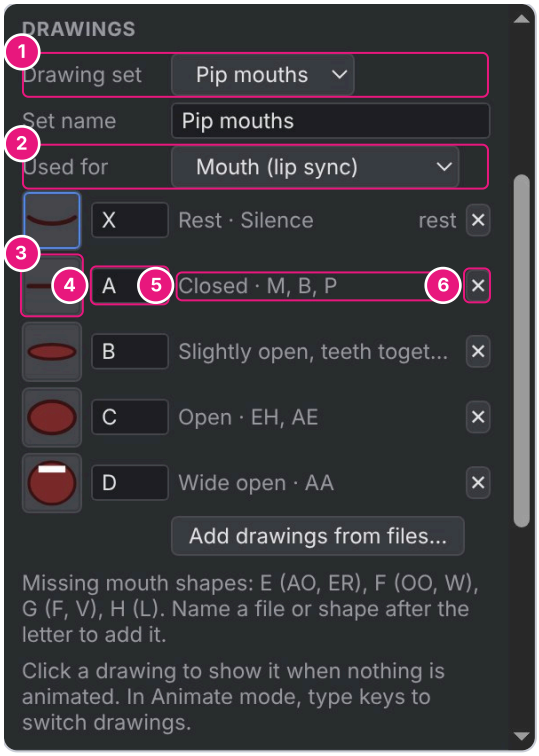
These are the classic cartoon mouth shapes. Draw one of each for your character (you can start with just A, C, D and X and add the rest later).

| | | |
|---|---|---|
| **A**<br>**Closed**<br>M, B, P ("mum", "big") | **B**<br>**Slightly open, teeth together**<br>K, S, T, EE and most consonants | **C**<br>**Open**<br>EH, AE ("bed", "cat") |
| **D**<br>**Wide open**<br>AA ("father") | **E**<br>**Slightly rounded**<br>AO, ER ("bird", "off") | **F**<br>**Puckered**<br>OO, W ("too", "we") |
| **G**<br>**Teeth on lower lip**<br>F, V ("fun", "van") | **H**<br>**Tongue raised**<br>L ("look") | **X**<br>**Rest**<br>Silence, between words |

## Making a mouth

1. In Build mode, draw each mouth shape. A mouth can be several shapes (lips, teeth, tongue); group each mouth's pieces with ⌘ + G.
2. Name each one after its letter (double-click the name in the Layers panel): **A**, **C**, **D**, **X**... Names like **rest**, **MBP**, **FV** or **mouth_D** work too.
3. Select all the mouth shapes and choose **Object → Make Switch Layer** (⇧ + ⌘ + M). They become one part called **Mouth**.
4. In the Layers panel, drag the Mouth onto the head, so it moves with the head. Use the Joints tool to place its joint in the middle of the mouth.

> **Using PNG pictures instead?** Name the files **A.png**, **D.png** and so on, select the Mouth, and click **Add drawings from files...** in Properties. Each file lands on its letter.

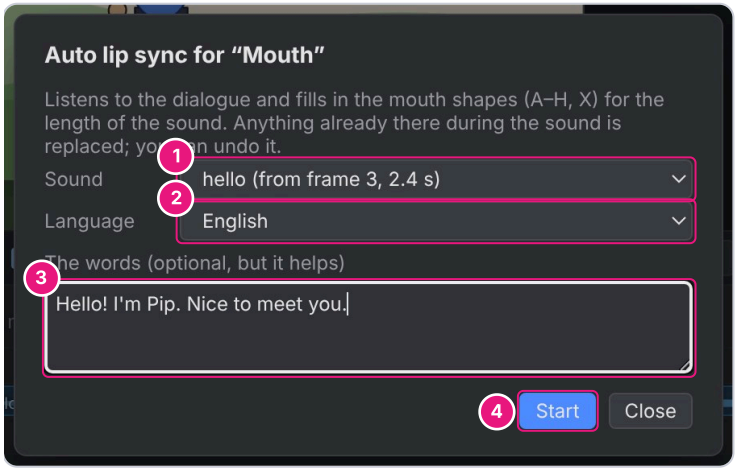



*The Mouth's drawings in Properties (Build mode).*

1. Which drawing set this switch layer uses. Swap sets to change a character's mouth art
2. Mouth (lip sync) or Other (eyes, hands...)
3. Each drawing: click the picture to make it the resting mouth
4. Its key: the letter you type. Rename it here
5. The sounds this mouth shape is for
6. Remove a drawing

Properties also lists any of the nine shapes that are still missing. Clicking a drawing in Build mode makes it the **rest** mouth: the one shown when nothing is animated.

## Automatic lip sync




*Auto lip sync, from the mouth palette.*

1. The dialogue to listen to
2. Its language: English, or Another language (matched by sound alone)
3. The words, typed exactly as spoken: optional, but it helps a lot
4. Start. A progress bar and Cancel appear while it listens

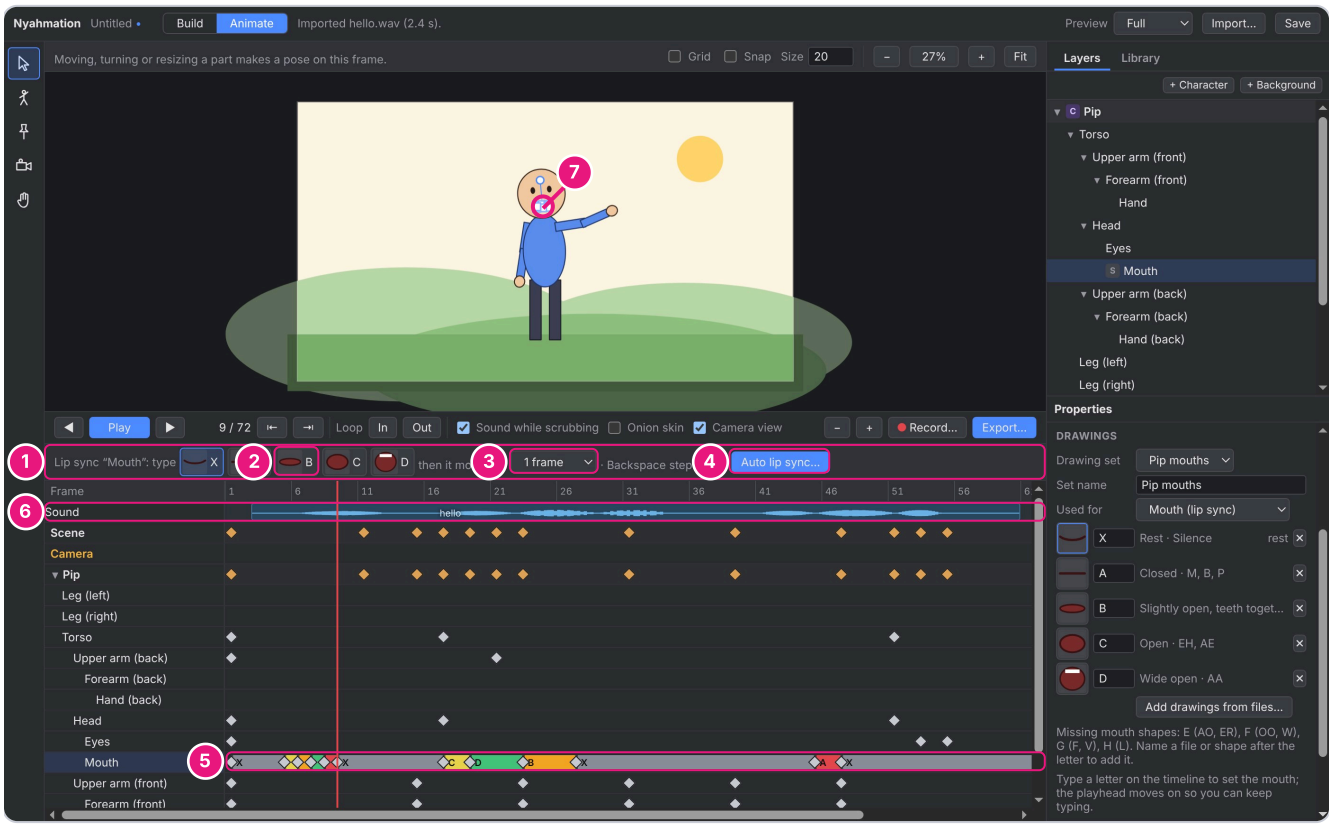
Nyahmation can listen to the dialogue and fill in the mouth shapes for you, using Rhubarb Lip Sync. It's a great first draft: it gets most shapes right, and you spend your time polishing instead of typing every letter.

1. Record or import the dialogue (Chapter 14) and switch to Animate mode.

2. Click the Mouth row's name on the timeline. The mouth palette appears; click **Auto lip sync...**
3. Pick the sound, the language, and if you can, type the words. Click **Start**.
4. After a few seconds to a minute, the Mouth row fills with shapes for the length of the sound. Play it, and fix any shape you don't like by typing its letter (below).

- Anything the mouth already did during the sound is replaced; before and after it is kept. It's one step to undo.
- If the mouth has no drawing for a shape, the nearest one it does have stands in, so an A-to-F set still moves.
- If the Auto lip sync window says it isn't installed yet, see Chapter 2.

## Lip syncing by hand




*Lip syncing: the mouth palette, the Mouth row's coloured blocks, and the dialogue on the Sound row.*

1. The mouth palette appears when a switch layer is selected in Animate mode
2. Type a letter (or click a picture) to show that mouth from this frame
3. After each letter the playhead moves on 1 or 2 frames
4. Auto lip sync...
5. The Mouth row: a coloured block for each mouth, from where it starts to the next change
6. The dialogue on the Sound row, to line the mouths up with
7. The mouth on this frame

1. Import the dialogue and switch to Animate mode.
2. Click the Mouth row's name on the timeline. The mouth palette appears above the timeline.

3. Go to the frame where the first word starts. Step with → and listen: you'll hear each frame.
4. Type the letter of the mouth shape for the sound you hear. The mouth changes on this frame and the playhead moves on one frame, so you can keep listening and typing.
5. If the sound hasn't changed, just press → to move on. A mouth stays until the next letter you type.
6. Made a mistake? Backspace steps back and clears that frame.
7. Press Space to watch it with the sound.

To retime one mouth, drag the left edge of its block on the Mouth row (⇧-drag moves the ones after it too).

> **Tips from animators.** Mouths usually look best a frame or two *before* the sound. To nudge a whole phrase, select its ◆ marks on the Mouth row and ⇧-drag. Don't try to show every letter: hit the big, visible sounds (M/B/P closing, wide vowels) and let the rest go.

> **Letters beat tool keys.** While the Mouth is selected in Animate mode, the letter keys set mouth shapes (so H sets the H mouth instead of choosing the Hand tool). Press Esc to deselect the Mouth and get the tool keys back.

## Other switch layers

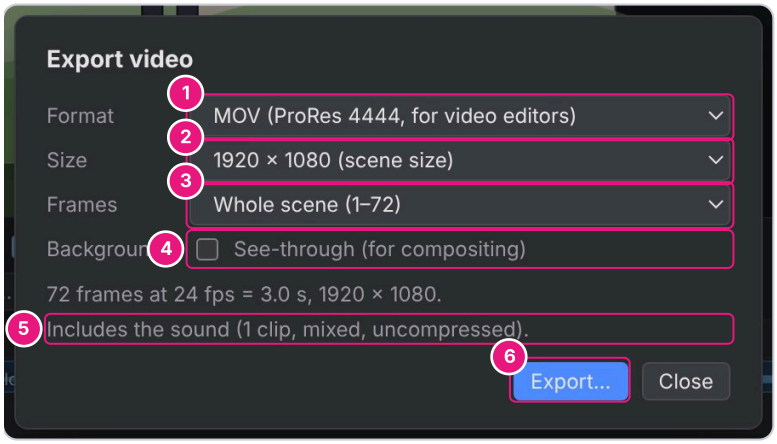
Switch layers aren't just for mouths: make one for eyes (open, half, closed, for blinks), hands (fist, open, pointing) or eyebrows. If the drawings aren't named after mouth letters, press 1–9 to pick them, or click them in the palette. The playhead doesn't move on for these.

A character's frames remember *letters*, not pictures. You can redraw a mouth, or swap in a different drawing set with **Drawing set**, and all the lip sync follows.

CHAPTER 16

# Exporting video

MP4, WebM, MOV and PNG sequences




*The Export window.*

1. Format: MP4, WebM, MOV or a PNG sequence
2. Size: the scene size, or 720p up to 4K
3. The whole scene, or just the loop range
4. See-through background (WebM, MOV and PNG)
5. Whether the sound is included
6. Choose where to save and start. A progress bar and Cancel appear while it works

1. Choose **File → Export Video...** (⌘ + E), or click **Export...** on the timeline.
2. Pick a format, a size and which frames.
3. Click **Export...** and choose where to save. A progress bar shows each frame and roughly how long is left; **Cancel** stops it.
4. When it's done, **Show in Finder** takes you to the video.

## Which format?

| FORMAT | BEST FOR |
|---|---|
| **MP4 video (H.264)** | Watching and sharing: plays on phones, computers, YouTube and messaging apps. Includes the sound. **If in doubt, pick this one.** |
| **MP4 video (H.265)** | The same quality in a smaller file. Plays on recent Macs, iPhones and most new devices; older computers may not play it. |
| **WebM video (VP9)** | Web pages. Can have a see-through background, to put a character over a web page or other footage. |
| **MOV (ProRes 4444)** | Video editing apps (Final Cut Pro, Premiere, DaVinci Resolve). Top quality, very big files, optional see-through background, and the sound uncompressed. |
| **PNG image sequence** | One picture per frame, for video editing and compositing apps. Can have a see-through background. The sound is saved beside the pictures as **soundtrack.wav**. |

- **Size:** the scene size, or 720p up to 4K. Bigger is sharper but takes longer and makes a bigger file. 1080p is a good choice.
- **Frames:** the whole scene, or only the loop range you set with <kbd>I</kbd> and <kbd>O</kbd>.
- **See-through** leaves out the scene's background colour and sky, so only the characters and scenery are drawn.
- Every frame is drawn at full quality with all its effects, whatever the Preview setting; it just takes a little longer on a slow computer. The video is shot through the camera.
- If a PNG picture in the scene will be shown bigger than its own pixels (including when the camera zooms in), the Export window lists it, because it may look soft. Pick a smaller size, or use a bigger picture.

CHAPTER 17

# Saving and files

Projects, autosave, the library folder, and safety

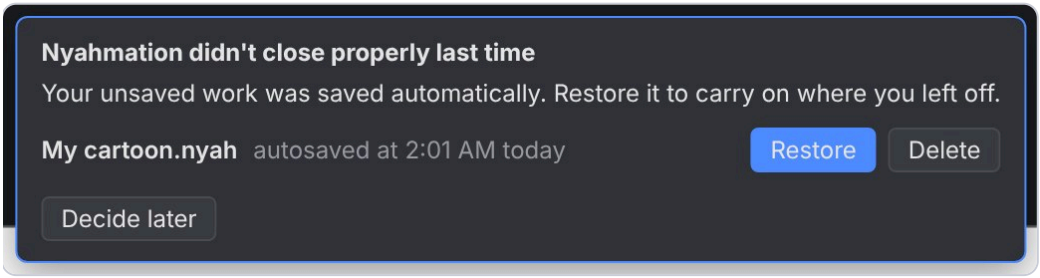
## Projects

A project is one scene, saved as one **.nyah** file. The file holds everything: the drawings, the animation, the camera, pictures and sounds, so you can copy it to another computer or back it up in one piece.

- **Save** (⌘ + S) and **Save As...** (⇧ + ⌘ + S). A dot after the project's name means there are unsaved changes; undoing back to how it was saved makes the dot go away.
- Nyahmation asks before closing, or opening another project, if there are unsaved changes.
- Saving is safe: the new file is written completely before it replaces the old one, so a crash can't damage your project.
- Double-click a .nyah file in Finder, drop it on Nyahmation's Dock icon or window, or pick it from **File → Open Recent**, to open it.
- If a file has been damaged (by a failing disk, say), Nyahmation says so instead of opening it, and your autosave or a backup can rescue the work.

## Autosave: a safety net

While a project has unsaved changes, Nyahmation quietly keeps a copy of it, about once a minute, in its own folder. It doesn't replace your file; it's a safety net under it, in case the computer crashes, the battery dies, or the app is forced to quit. When you save, open another project or discard your changes, the copy is deleted.




*What you see the next time you open Nyahmation after a crash.*

If Nyahmation didn't close properly last time, it says so when it starts, with the name of the work and when it was last copied:

| BUTTON | WHAT HAPPENS |
|---|---|
| **Restore** | Opens the work as unsaved changes to its original file. Press Save to put it back where it came from. |
| **Delete** | Throws the copy away. |

| BUTTON | WHAT HAPPENS |
| --- | --- |
| **Decide later** | Keeps the copy and asks again the next time Nyahmation starts. |

> **Still save often.** Autosave keeps at most a minute's work at risk, but only ⌘ + S updates your own file.

## The library folder

Library items are **.nyahitem** files in **Documents/Nyahmation Library**. They are ordinary files: you can back them up, share them, or organise them in folders.

## Undo

Almost everything can be undone with ⌘ + Z and redone with ⇧ + ⌘ + Z, including drawing, posing, retiming, camera moves, effects, pastes and automatic lip sync.

CHAPTER 18

# Keyboard shortcuts

Every key in one place

## TOOLS

| | |
|---|---|
| Select | V |
| Points | A |
| Joints | J |
| Pose | K |
| Pen (Build) / Pin (Animate) | P |
| Camera (Animate) | C |
| Rectangle | M |
| Ellipse | L |
| Polygon | Y |
| Star | S |
| Line | \ |
| Hand | H or hold Space |

## BUILD MODE

| | |
|---|---|
| Nudge | Arrow keys (⇧ = 10 px) |
| Delete selection | Delete |
| Deselect / stop editing points | Esc |
| Select parent | ⇧ + Enter |
| Finish a pen line | Enter |

## ANIMATE MODE

| | |
|---|---|
| Play / pause | Space |
| Previous / next frame | ← → |
| Previous / next pose | ⇧ + ← ⇧ + → |
| First / last frame | Home End |
| Loop in / out | I O |
| Delete selected poses or sound | Delete |
| Deselect everything | Esc |
| Copy / paste a pose | ⌘ + C ⌘ + V |
| Mirror Pose | ⌥ + ⌘ + M |
| Swap Sides | ⌥ + ⇧ + ⌘ + M |
| Change draw order from here | ⌘ + [ ⌘ + ] |

## LIP SYNC (MOUTH SELECTED)

| | |
|---|---|
| Set a mouth shape | A – H, X |
| Pick drawing 1–9 | 1 – 9 |
| Step back and clear | Backspace |

## TIMELINE DRAGGING

| | |
|---|---|
| Retime a pose | drag ◆ |
| Retime and shift the rest | ⇧ drag |

| BUILD MODE | |
|---|---|
| **Cut / copy / paste parts** | ⌘ + X ⌘ + C ⌘ + V |
| **Union / Subtract** | ⌥ + ⌘ + U ⌥ + ⌘ + S |
| **Intersect / Exclude** | ⌥ + ⌘ + K ⌥ + ⌘ + E |

| CAMERA TOOL (ANIMATE) | |
|---|---|
| **Pan** | drag |
| **Zoom** | ⌥ drag up/down |
| **Turn** | ⇧ drag |

| TIMELINE DRAGGING | |
|---|---|
| **Copy a pose (hold)** | ⌥ drag |
| **Select more poses** | ⇧ click |
| **Retime a mouth** | drag its block's left edge |

On a MacBook keyboard, Home is Fn + ← and End is Fn + →. The other menu shortcuts ( ⌘ + S, ⌘ + Z and so on) are listed in Chapter 3.

CHAPTER 19

# Troubleshooting

When something goes wrong

| PROBLEM | WHAT TO DO |
|---|---|
| **Nothing happens when I draw** | The layer may be locked or hidden (check the padlock and eye in the Layers panel), or you may be in Animate mode, where the drawing tools aren't available. |
| **Dragging the hand tips the whole body over** | Select the character's layer and click **Mark branch joints as chain roots**, or double-click the shoulder joint with the Joints tool to make it a chain root. |
| **An arm bends the wrong way** | Select the forearm and change **Bends** in its joint settings, or set joint limits. |
| **A part spins around the wrong point** | Its joint is in the wrong place. Move it with the Joints tool (`J`). |
| **Letter keys change tools instead of mouths** | Select the Mouth first (click its row on the timeline), and make sure you're in Animate mode. |
| **Typing a letter doesn't change the mouth** | The mouth set has no drawing with that letter. Check the Mouth's drawings in Properties. |
| **Auto lip sync says it isn't installed** | In Terminal, in the Nyahmation folder, run `npm run setup` (it needs the internet, about 90 MB), then restart Nyahmation (or rebuild the app with `npm run dist:mac`). |
| **Auto lip sync gets the mouths wrong** | Type the words in the dialog, exactly as spoken; check the language; use a clear recording without music under it. Then fix the rest by typing letters. |
| **I can't hear anything** | Check the Mac's volume, that the sound isn't muted in its properties, and that **Sound while scrubbing** is ticked. |
| **Playback is jerky, or shows fewer fps in orange** | Choose a lower **Preview** quality in the top bar, or use **View → Measure Preview Speed**. Blur and large glows are the slowest effects. The export isn't affected. |
| **The character looks different on screen and in the video** | Check **Camera view** is ticked: with it off you see the whole stage, not what the camera films. **View on ones** also shows smoother motion than a character on twos will export with. |
| **The feet slide while walking** | Pin them (Chapter 9). With a cycle, tick **Keep moving** so the pins move on with each step. |

| PROBLEM | WHAT TO DO |
| --- | --- |
| **A cycle jumps at every repeat** | The cycle's last frame should match its first: copy the pose on frame *From* and paste it on frame *To*. |
| **Mirror Pose leaves a part unchanged** | It pairs parts by name. Name them with left/right, front/back or L/R ("Arm (left)" and "Arm (right)"). |
| **The pupil pokes out of the eye** | Put the pupil inside the eye shape in the Layers panel, and tick **Clip** on the eye. |
| **A picture looks blurry in the video** | It's being shown bigger than its own pixels, perhaps because the camera zooms in. Export at a smaller size, or import a bigger PNG. Vector drawings are always sharp. |
| **The H.265 video won't play** | Some older computers and apps can't play H.265. Export as MP4 (H.264) instead. |
| **Nyahmation closed unexpectedly** | Open it again and click **Restore** (Chapter 17). |
| **macOS says the app can't be opened** | System Settings → Privacy & Security → **Open Anyway** (see Chapter 2). |
| **"Electron uninstall" when running npm run dev** | Run `npm run setup` in the Nyahmation folder; it downloads the parts that newer versions of npm skip. |

CHAPTER 20

# Glossary

Animation words used in this manual

| WORD | MEANING |
|---|---|
| **Blend mode** | How a part mixes with what's behind it: covering it, darkening it (Multiply) or lightening it (Screen, Add). |
| **Camera** | What films the stage. Moving it changes the picture, never the characters. |
| **Chain root** | A joint where posing stops. Dragging a hand turns the arm, but not past a chain root at the shoulder. |
| **Clip** | Showing the parts inside a shape only where the shape is, like pupils inside an eye. |
| **Contact (pose)** | The moment in a walk when the front heel touches the ground. |
| **Cycle** | A stretch of animation that repeats, such as one step of a walk. |
| **Depth** | How far away a layer is, which decides how much it moves when the camera does. |
| **Easing** | How a movement speeds up and slows down between two poses. |
| **Effect** | A finishing touch drawn on a part or layer as a whole: shadow, glow, blur, haze. |
| **Frame** | One picture of the animation. At 24 frames per second, 24 frames make one second of video. |
| **Frame rate (fps)** | How many frames are shown each second. |
| **Gradient** | A fill that fades from one colour to another. |
| **Hold** | A pause: the same pose on two frames, so nothing moves between them. |
| **IK (inverse kinematics)** | Dragging the end of a limb (a hand) and letting the joints above it (elbow, shoulder) work out how to bend. |
| **In-betweens** | The frames between two poses, which Nyahmation fills in for you. |
| **Joint** | The point a part turns around, where it attaches to its parent. |
| **Layer** | A level of the scene: a character or a background. |
| **Lip sync** | Matching a character's mouth shapes to the words it says. |

| WORD | MEANING |
| --- | --- |
| **Loop range** | A stretch of frames that playback repeats, set with `I` and `O`. |
| **On twos** | Showing each pose for two frames, for a hand-drawn feel. |
| **Onion skin** | Faint copies of the frames before and after, to see a movement's path. |
| **Overshoot** | Moving a little past a pose and settling back, for a lively, springy feel. |
| **Parallax** | Near things seeming to move faster than far things as the camera moves. |
| **Parent / child** | A child part is inside its parent and moves with it (the hand is a child of the forearm). |
| **Passing (pose)** | The moment in a walk when one leg passes the other. |
| **Pin** | Fixing a part to a spot on the stage from a certain frame, so it stays put while the rest moves. |
| **Pose** | A position of a part (or the camera) recorded on a frame, shown as ◆ on the timeline. |
| **Rest pose** | How a character looks when nothing is animated; set in Build mode. |
| **Rig** | A character's joints and part tree, which make it posable like a puppet. |
| **Scrubbing** | Dragging the playhead back and forth to see (and hear) the frames. |
| **Stage** | The world the characters and scenery stand in, which the camera films. |
| **Swatch** | A saved colour in the project's palette. |
| **Switch layer** | A part that shows one drawing at a time from a set, such as mouth shapes. |
| **Vector** | Artwork made of points and curves, which stays sharp at any size. |